

Endpoint competition in a Galton board with memory: uniform thresholds, boundary conditions, and simplex modes

Problem 131 and its persistent-walk extensions

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Abstract

A ball on a Galton board bounces left or right with equal probability at the first peg and afterwards repeats its previous direction with probability p . Problem 131 of Kagey's Open Problem Collection asks for the distribution over the bins, for the p at which the middle bin is as likely as the extreme bins, for the relation to lattice walks and to OEIS A035002 and A348595, for other board geometries, and for the normal limit and its variance.

The model is the correlated random walk of Goldstein and Gillis, and the exact distribution is a run count going back to Wald and Wolfowitz; we record it with a short proof, a rational generating function and a three-term Pascal-type recurrence. We prove by explicit bijections that Kagey's numerator table for $p = 2/3$ is the rook-path array A035002 and that his table for $p = 1/3$ is the symmetric array whose lower half is A348595. The bin index after N bounces has variance $Np/(4(1-p)) - (2p-1)(1-(2p-1)^N)/(8(1-p)^2)$ and is asymptotically normal for $0 < p < 1$. In every row with $N \geq 2$ there is exactly one $p = p_N$ at which the middle equals the extremes; p_N increases strictly from $p_2 = 2/3$, $p_3 = 1/\sqrt{2}$ to 1. A uniform Bessel approximation proves $N(1-p_N) = \log N - \frac{1}{2} \log \log N + \frac{1}{2} \log(\pi/8) + o(1)$ and yields a further inverse-logarithm correction. We extend the relative Bessel bound to every interior bin, with error at most $Nu^2/2 + u$, where $u = (1-p)/p$. This yields a threshold expansion uniform over all diverging distances from an edge, matched to a Laguerre-polynomial law at fixed distance. For periodic Ising boundaries we obtain the corresponding estimate, a rigorous refinement of a previously studied finite-size crossing, and an explicit $\log 2$ boundary shift. Both boundary conditions have the spatial crossing profile e^{t-2y^2} . For the symmetric d -direction board, a positive Bessel integral gives sharp crossings for every balanced simplex face and a global-mode transition in the logarithmic window. An estimate uniform over all positive compositions then resolves a rare-coordinate crossover at $N/(\log N)^2$. We also determine the spatial volume of superlevel bins and prove that weak occupation fields can stabilize every intermediate support size. Near a vertex, a product of finite Laguerre polynomials determines the crossing and its first correction, which depends on the number and multiplicities of the rare colors. The leading Laguerre-product root remains valid uniformly for arbitrary growing rare counts whose total is $A = o(b)$, where b is the dominant count, with relative root error $O_\ell(\sqrt{(A+1)/b})$ for a fixed number ℓ of rare colors. This root approximation can hold even when the corresponding relative probability approximation fails by an unbounded factor. Exact laws, ordinary normal limits, and fixed-scale Bessel distributions are established theory; the proposed contributions are the quantitative moving-parameter refinements.

1 Introduction

Problem 131 of Kagey’s collection [1] reads as follows. Model a Plinko or Galton board by assuming that a ball (1) is equally likely to bounce in either direction off the first peg, and (2) thereafter bounces in the same direction it previously bounced with probability p . So for $p = 1$ the ball always repeats its direction, for $p = \frac{1}{2}$ the bounces are independent, and for $p = 0$ the ball alternates. The page shows “numerator tables” for $p = \frac{1}{3}, \frac{2}{3}, \frac{5}{6}$ and the probability mass functions of the tenth row, and asks: *for an arbitrary $p \in [0, 1]$ and a triangle with n rows, what is the distribution of balls in bin k ?* Four related questions follow.

- R1.** What is the least/greatest value of p such that at the $(2n)$ -th row the middle entry equals the extremes? How about the $(2n + 1)$ -st row?
- R2.** What changes on a different geometry (a cylinder, a tetrahedron)?
- R3.** How does this relate to lattice walks? (See A348595.)
- R4.** As $n \rightarrow \infty$, does the distribution converge to a normal distribution? If so, what is the variance as a function of p ?

The references given are A035002 (numerators for $p = \frac{2}{3}$) and A348595 (“related” to $p = \frac{1}{3}$).

Conventions. We index by the number $N \geq 0$ of bounces. After N bounces the ball is in one of $N + 1$ bins, and bin $k \in \{0, \dots, N\}$ means k bounces to the right. Kagey’s n -th row has n entries, so it is our $N = n - 1$; his tenth-row histograms have ten bars. We write $q = 1 - p$, $\rho = 2p - 1$, and $P_N(k)$ for the probability of bin k after N bounces. For $p = \alpha/(\alpha + \beta)$ in lowest terms, the entries of Kagey’s tables are

$$n_N(k) = 2(\alpha + \beta)^{N-1} P_N(k) \quad (N \geq 1), \quad n_0(0) = 1,$$

not reduced further. This normalization reproduces all 3 tables of the figure, rows 1 to 9.

What is answered. The main question is answered by Theorem 2.1 and Proposition 2.2. R3 and the two OEIS references are settled by Theorems 3.2 and 3.3. R4 is answered by Proposition 4.1 and Theorem 4.2: the limit is normal for $0 < p < 1$ and the variance of the bin index is asymptotically $Np/(4(1 - p))$. R1 is answered by Theorem 5.1 and the paragraph after it: each row has exactly one solution p_N , the least values among nontrivial rows are $\frac{2}{3}$ (odd rows) and $1/\sqrt{2}$ (even rows), and there is no greatest value. Theorem 5.2 proves a quantitative Bessel approximation and its second-order expansion. Proposition 5.3 gives the adjacent-bin crossing. For R2, Remark 4.3 treats the cylinder, and Propositions 2.3 and 2.4 and Theorem 4.5 give an exact law and a normal limit for the explicit symmetric tetrahedral model defined below. Other geometric transition rules can define different processes.

The threshold analysis is developed in Sections 6, 7 and 8. It extends the binary comparison to all edge distances and compares free with periodic spin boundaries. In the specified multidirectional model it also determines the sharp balanced-face crossings and the global modes in their critical window. The complete finite- N multistate mode classification is not asserted: a four-step, three-direction example has its maxima on proper edges. Sections 9 and 11 extend the analysis to rare coordinates, spatial superlevel volumes, and weak-field face phases. Section 10 completes the leading near-vertex crossing law uniformly over all rare-count totals $A = o(b)$, including arbitrarily unequal counts.

Literature. The position of the ball is the *correlated* (or *persistent*) random walk, studied since Goldstein [5] and Gillis [6]. Renshaw and Henderson [7] treat exactly the walk that continues with probability p and reverses with probability $1 - p$; according to its abstract that paper derives exact n -step transition probabilities and limiting distributions. We have consulted only the abstract, and our proofs are self-contained. The count of binary words by number of runs, which is all that Theorem 2.1 needs, is the classical computation of Wald and Wolfowitz [8]. Hence, the exact law and the normal limit should not be regarded as new. Dekking and Kong [10] give the Markov-binomial law, variance, interior log-concavity, and local modality classification; their parameters $a = b = 1 - p$ and symmetric initial law give our model. Local modality is distinct from equality of central and endpoint heights. Herrmann and Vallois [11] give the classical telegraph limit with its I_0, I_1 density. The quantitative estimate here treats the moving regime $Nu = O(\log N)$, which does not follow from fixed-scale weak convergence alone.

The d -direction model belongs to Markov multinomial theory [12, 13]; we have not obtained the original Wang–Yang paper and make no novelty claim for the finite sum or normal limit given here. The OEIS entries [2, 3] already record the relevant generating functions, and Kagey already indicates their connections; our bijections explain these identifications.

The Ising literature gives a closer antecedent than the initial search identified. Antal, Droz and Rácz [14] derive fixed-scale magnetization distributions under several boundary conditions. Dantchev and Rudnick [15] give exact free and periodic fixed-magnetization formulas and periodic Bessel scaling. Stepanyan et al. [16] explicitly define the periodic endpoint–center and endpoint–adjacent crossings and report a finite-size approximation to the former. Thus the crossing question itself is already present in the literature. The refinements here control the discrete error while the persistence parameter moves with N , invert the resulting comparison quantitatively, and determine a spatial superlevel-set window. For the multistate extension, general continuous-time occupation densities on visited simplex faces are also known [17]; the discrete moving-scale comparison proved below is a different assertion. The square-root composition quantity $A^2 - 1$, where $A = \sum_i \sqrt{\alpha_i}$, is also classical: for the complete-graph generator Q with off-diagonal entries one, the symmetric occupation rate is $I_Q(\alpha) = d - A^2$. Thus $A^2 - 1$ is the difference between the vertex rate and the rate at α , not a new rate function. The results proposed here concern uniform discrete relative errors and their boundary-crossing and modal consequences.

No matching theorem for the full quantitative package was located in the inspected sources. This bounded search does not establish priority. The full Renshaw–Henderson and Wang–Yang papers remain to be checked. Library access allowed us to inspect Čekanavičius and Mikalauskas [18]: Theorem 1.1 assumes $p, \bar{q} \leq 1/2$, where p is the repeat probability in state one. It therefore does not cover our symmetric $p \rightarrow 1$ regime; the paragraph following Corollary 1.1 explicitly leaves a high-persistence range unresolved there. Their separate large-deviation paper [19] was also inspected: Theorems 1.1–1.2 require $p \leq 1/50$ and Theorem 1.3 requires $p \leq \bar{q}$. These hypotheses likewise exclude the symmetric persistence limit studied here.

2 The exact distribution

A run of the experiment with $N \geq 1$ bounces is a word $w \in \{L, R\}^N$. Let $\text{chg}(w)$ be the number of indices $i < N$ with $w_i \neq w_{i+1}$ and $\text{rep}(w) = N - 1 - \text{chg}(w)$. By the definition of the model,

$$\mathbb{P}(w) = \frac{1}{2} p^{\text{rep}(w)} q^{\text{chg}(w)}. \quad (1)$$

Theorem 2.1 (bin probabilities). *Let $N \geq 1$. Then $P_N(0) = P_N(N) = \frac{1}{2}p^{N-1}$, and for $1 \leq k \leq N-1$*

$$P_N(k) = \frac{1}{2} \sum_{j=1}^{N-1} W(N, k, j) p^{N-1-j} q^j,$$

where, writing $a = k$ and $b = N - k$,

$$W(N, k, 2i-1) = 2 \binom{a-1}{i-1} \binom{b-1}{i-1}, \quad W(N, k, 2i) = \binom{a-1}{i} \binom{b-1}{i-1} + \binom{a-1}{i-1} \binom{b-1}{i}$$

is the number of words with a letters R , b letters L and exactly j direction changes.

Proof. By (1) it suffices to count words. A word with j changes consists of $j+1$ maximal runs, alternately of R 's and of L 's. If $k \in \{0, N\}$ there is one word and it has no change. Let $a, b \geq 1$. If $j+1 = 2i$ there are i runs of each letter and the first letter is free; the run lengths form a composition of a into i parts and a composition of b into i parts, and there are $\binom{a-1}{i-1} \binom{b-1}{i-1}$ such pairs. If $j+1 = 2i+1$ the word starts and ends with the same letter; starting with R gives $i+1$ runs of R and i runs of L , that is $\binom{a-1}{i} \binom{b-1}{i-1}$ words, and starting with L gives the other summand. $\square$

For instance $P_2 = (\frac{p}{2}, q, \frac{p}{2})$, $P_3(1) = P_3(2) = \frac{1}{2}(1-p^2)$ and $P_4(2) = (1-p)(1-p+p^2)$.

Proposition 2.2 (recurrence and generating function). *Let $R_N(k)$ and $L_N(k)$ be the probabilities of being in bin k after N bounces with last bounce R , respectively L . Then $R_1(1) = L_1(0) = \frac{1}{2}$ and*

$$R_N(k) = p R_{N-1}(k-1) + q L_{N-1}(k-1), \quad L_N(k) = p L_{N-1}(k) + q R_{N-1}(k). \quad (2)$$

Consequently

$$\sum_{N \geq 0} \sum_{k=0}^N P_N(k) x^k t^N = \frac{1 - (p - \frac{1}{2})(1+x)t}{1 - p(1+x)t + (2p-1)xt^2}, \quad (3)$$

and for all $N \geq 2$ and all k (with $P_M(k) = 0$ outside $0 \leq k \leq M$)

$$P_N(k) = p(P_{N-1}(k) + P_{N-1}(k-1)) - (2p-1)P_{N-2}(k-1). \quad (4)$$

Proof. The recursion (2) follows by conditioning on the last two bounces. Put $F_R = \sum R_N(k) x^k t^N$ and $F_L = \sum L_N(k) x^k t^N$. Then (2) and the initial values give $F_R = xt(\frac{1}{2} + pF_R + qF_L)$ and $F_L = t(\frac{1}{2} + pF_L + qF_R)$. The determinant of this linear system is $(1 - pxt)(1 - pt) - q^2xt^2 = 1 - p(1+x)t + (p^2 - q^2)xt^2$, and $p^2 - q^2 = 2p - 1$. Cramer's rule gives $F_R = \frac{1}{2}xt(1 - pt)/\Delta$ and $F_L = \frac{1}{2}t(1 - pxt)/\Delta$ with Δ the determinant, and $1 + F_R + F_L$ simplifies to (3). Multiplying (3) by Δ and reading off the coefficient of $x^k t^N$ for $N \geq 2$ gives (4). $\square$

At $p = \frac{1}{2}$ the last term of (4) vanishes and (4) is Pascal's rule with weights $\frac{1}{2}$. In terms of numerators, (4) reads $n_N(k) = \alpha(n_{N-1}(k) + n_{N-1}(k-1)) - (\alpha^2 - \beta^2)n_{N-2}(k-1)$ for $N \geq 3$.

The same computation works for a board on which each peg offers $d \geq 2$ directions: the first bounce is uniform, and afterwards the ball repeats its direction with probability p and takes each other direction with probability $q/(d-1)$. For $d = 3$ this is a tetrahedral pile of pegs, with the bins of layer N indexed by (k_1, k_2, k_3) , $k_1 + k_2 + k_3 = N$.

Proposition 2.3 (d directions). *Let $P_N(k_1, \dots, k_d)$ be the probability that direction i is taken k_i times. Put $\sigma = p - \frac{q}{d-1} = \frac{dp-1}{d-1}$ and $A = \sum_{i=1}^d \frac{x_i t}{1 - \sigma x_i t}$. Then*

$$\sum_{N \geq 0} \sum_{k_1 + \dots + k_d = N} P_N(k_1, \dots, k_d) x_1^{k_1} \dots x_d^{k_d} t^N = 1 + \frac{A/d}{1 - \frac{q}{d-1} A}.$$

Proof. Let F_i be the generating function of the runs whose last direction is i , and $S = \sum_i F_i$. Conditioning on the last bounce, $F_i = x_i t \left(\frac{1}{d} + p F_i + \frac{q}{d-1} (S - F_i) \right)$, that is $F_i(1 - \sigma x_i t) = x_i t \left(\frac{1}{d} + \frac{q}{d-1} S \right)$. Summing over i gives $S = A \left(\frac{1}{d} + \frac{q}{d-1} S \right)$, and the generating function is $1 + S$. $\square$

For $d = 2$, $x_1 = x$, $x_2 = 1$ this reduces to (3), since then $\sigma = \rho$.

Proposition 2.4 (finite bin probabilities in d directions). *Put $r = q/(d-1)$ and $\sigma = p - r$. For $N \geq 1$, let $k = (k_1, \dots, k_d)$ have nonnegative integer coordinates summing to N , and put $H = \{i : k_i > 0\}$. Sum over $m_i = 0$ for $i \notin H$ and $1 \leq m_i \leq k_i$ for $i \in H$, and set $M = \sum_i m_i$. Then*

$$P_N(k) = \frac{1}{d} \sum_m \frac{M!}{\prod_i m_i!} r^{M-1} \sigma^{N-M} \prod_{i \in H} \binom{k_i - 1}{m_i - 1}.$$

All exponents are nonnegative, and $0^0 = 1$. Set $P_0(0, \dots, 0) = 1$ separately.

Proof. With $z_i = x_i t$, Proposition 2.3 becomes the formal power series

$$1 + \frac{1}{d} \sum_{M \geq 1} r^{M-1} \left(\sum_i \frac{z_i}{1 - \sigma z_i} \right)^M.$$

Expand by the multinomial theorem and use, for $k \geq m \geq 1$,

$$[z^k] \left(\frac{z}{1 - \sigma z} \right)^m = \binom{k-1}{m-1} \sigma^{k-m}.$$

For $m = 0$ only $k = 0$ contributes. Multiplying the coefficients proves the formula. $\square$

At $d = 3$ this is a triple sum, with the support convention covering edge bins. At $p = 1/d$, $\sigma = 0$ and only $M = N$ survives, giving the ordinary multinomial law. At $p = 1$, only $M = 1$ survives, giving mass $1/d$ at each Ne_i . The formula is valid also at $p = 0$; for $\sigma < 0$ its terms can cancel, so dynamic programming or exact arithmetic is preferable numerically. This geometry assumes fixed global directions and uniform switching to the other directions; a different physical boundary or orientation rule need not have this law.

3 The numerator tables and lattice paths

Proposition 3.1. *Let $p = \alpha/(\alpha + \beta)$ with integers $\alpha, \beta \geq 0$, not both zero. For $N \geq 1$,*

$$n_N(k) = \sum_w \alpha^{\text{rep}(w)} \beta^{\text{chg}(w)},$$

the sum over the words with k letters R and $N - k$ letters L . In particular the numerators are integers, and

$$\sum_{a, b \geq 0} n_{a+b}(a) x^a y^b = \frac{1 - (\alpha - 1)(x + y) + (\alpha - \beta)(\alpha + \beta - 2)xy}{1 - \alpha(x + y) + (\alpha^2 - \beta^2)xy}.$$

Proof. Multiply (1) by $2(\alpha + \beta)^{N-1}$. For the generating function let F_R, F_L collect the weighted words ending in R , in L ; then $F_R = x(1 + \alpha F_R + \beta F_L)$ and $F_L = y(1 + \alpha F_L + \beta F_R)$, and solving as in Proposition 2.2 gives the stated value of $1 + F_R + F_L$. $\square$

For $(\alpha, \beta) = (2, 1)$ the generating function is $(1 - x)(1 - y)/(1 - 2x - 2y + 3xy)$, and for $(\alpha, \beta) = (1, 2)$ it is $(1 - xy)/(1 - x - y - 3xy)$. These are the generating functions recorded in A035002 (by Kruchinin and by Somos) and in A348595 (by Mathar) [2, 3], which already identifies the tables. We prove the identifications directly from the definitions of the two sequences, which shows where the weights 2 come from.

Theorem 3.2 ($p = \frac{2}{3}$ and rook paths). *Let $T(a, b)$ be the number of ways to move a rook from $(0, 0)$ to (a, b) by moves that increase one coordinate by a positive amount. Then for $p = \frac{2}{3}$, $n_N(k) = T(k, N - k)$ for all $N \geq 0$ and $0 \leq k \leq N$. In the indexing of A035002, whose array starts at $(1, 1)$, this is the entry $(k + 1, N - k + 1)$, so Kagey's $p = \frac{2}{3}$ table read by rows is A035002 read by antidiagonals.*

Proof. A035002 is defined by the rule that each entry is the sum of all entries to its left and all entries above it, which is the recursion for T obtained by removing the last rook move (the OEIS entry records this interpretation). Given a rook path to (a, b) with $a + b = N \geq 1$, replace each move of length ℓ to the right by R^ℓ and each move of length ℓ upward by L^ℓ . This gives a word w with a letters R and b letters L , together with the set of positions $i < N$ at which one move ends and the next begins. Such a cut is forced wherever $w_i \neq w_{i+1}$ and is optional wherever $w_i = w_{i+1}$, and every choice of optional cuts comes from exactly one rook path. Hence, $T(a, b) = \sum_w 2^{\text{rep}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (2, 1)$ by Proposition 3.1. $\square$

Theorem 3.3 ($p = \frac{1}{3}$ and Mathar's blocked walks). *Call a lattice point (x, y) blocked if $x \equiv y \equiv 1$ or $x \equiv y \equiv 2 \pmod{3}$. Let $M(a, b)$ be the number of walks from $(0, 0)$ to $(3a, 3b)$ with unit steps right and up that visit no blocked point. Then for $p = \frac{1}{3}$, $n_N(k) = M(k, N - k)$. The array M is symmetric, and A348595 lists its lower triangle $M(n, k)$, $0 \leq k \leq n$; this is the sense in which that entry is "related" to the $p = \frac{1}{3}$ table.*

Proof. Reduce coordinates modulo 3. The allowed residues are $(0, 0)$, $(1, 0)$, $(2, 0)$, $(2, 1)$, $(0, 1)$, $(0, 2)$, $(1, 2)$. Call a point with residue $(0, 0)$ a *corner*, and let (X, Y) count how many times the walk has crossed a vertical line $x \equiv 0$, respectively a horizontal line $y \equiv 0$, by a step landing on it. From a corner the walk steps right or up. After a right step it is at residue $(1, 0)$, and the step up is blocked, so it continues to $(2, 0)$; call this state A. From A a right step lands on a corner and increases X ; an up step leads to $(2, 1)$, where up is blocked, and then right to residue $(0, 1)$, which increases X , and then (right being blocked) up to $(0, 2)$; call this state B. By symmetry, from a corner an up step leads to B without changing (X, Y) , and from B the walk either reaches a corner or reaches A, increasing Y by one in both cases. Hence, every walk from A emits one letter R (an increase of X) and then is at a corner or at B; every walk from B emits one letter L and then is at a corner or at A; and a corner leads to A or to B without emitting anything. A walk from the corner $(0, 0)$ to the corner $(3a, 3b)$ therefore determines a word w with a letters R and b letters L : the letter w_i is emitted from A if it is R and from B if it is L . Between w_i and w_{i+1} the walk goes from one emitting state to the next. If $w_i = w_{i+1}$ it must pass through a corner (one way). If $w_i \neq w_{i+1}$ it may pass through a corner or go directly (two ways). The start and the end are forced. Hence $M(a, b) = \sum_w 2^{\text{chg}(w)}$, which is $n_N(a)$ for $(\alpha, \beta) = (1, 2)$. In Mathar's note [4] attached to A348595 the "squares" of the sequence name are the lattice sites (x, y) themselves, as his recurrences show, so M is his count; a direct count of the walks also reproduces the published terms (Section 12). $\square$

For general $p = \alpha/(\alpha + \beta)$, Proposition 3.1 says that $n_N(k)$ counts monotone lattice paths from $(0, 0)$ to $(k, N - k)$ in which each straight continuation is colored in one of α ways and each corner in one of β ways. This is the general answer to R3.

4 Variance and the normal limit

Write $e_i = +1$ if the i -th bounce is R and $e_i = -1$ otherwise, $X_N = e_1 + \dots + e_N$, and $K_N = (N + X_N)/2$ for the bin index. Then $e_{i+1} = e_i s_i$, where $s_1, s_2, \dots$ are independent of each other and of e_1 , with $\mathbb{P}(s_i = 1) = p$ and $\mathbb{P}(s_i = -1) = q$, so $\mathbb{E}s_i = \rho$.

Proposition 4.1 (mean and variance). $\mathbb{E}K_N = N/2$, and for $p < 1$

$$\text{Var } K_N = \frac{1}{4} \left(N \frac{1 + \rho}{1 - \rho} - \frac{2\rho(1 - \rho^N)}{(1 - \rho)^2} \right) = \frac{Np}{4(1 - p)} - \frac{(2p - 1)(1 - (2p - 1)^N)}{8(1 - p)^2}.$$

For $p = 1$, $\text{Var } K_N = N^2/4$.

Proof. Note that $\mathbb{E}e_1 = 0$ and $e_{i+m} = e_i s_i \dots s_{i+m-1}$, so $\mathbb{E}e_i = 0$ and $\mathbb{E}e_i e_{i+m} = \rho^m$. Hence, $\text{Var } X_N = N + 2 \sum_{m=1}^{N-1} (N - m) \rho^m$, and summing the two geometric series gives the first form; the second uses $(1 + \rho)/(1 - \rho) = p/q$ and $1 - \rho = 2q$. $\square$

Theorem 4.2 (normal limit). Let $0 < p < 1$. Then $(K_N - N/2)/\sqrt{N}$ converges in distribution to the normal law with mean 0 and variance $p/(4(1 - p))$.

Proof. Let $\varphi_N(\theta) = \mathbb{E}e^{i\theta X_N}$ and let v_N be the column vector with entries $\mathbb{E}[e^{i\theta X_N}; e_N = +1]$ and $\mathbb{E}[e^{i\theta X_N}; e_N = -1]$. By (2), $v_{N+1} = M v_N$ with

$$M = \begin{pmatrix} p e^{i\theta} & q e^{i\theta} \\ q e^{-i\theta} & p e^{-i\theta} \end{pmatrix}, \quad \text{tr } M = 2p \cos \theta, \quad \det M = p^2 - q^2 = \rho.$$

The eigenvalues are $\lambda_{\pm}(\theta) = p \cos \theta \pm \sqrt{q^2 - p^2 \sin^2 \theta}$, using $p^2 - \rho = q^2$. For $|\sin \theta| < q/p$ they are real and distinct, so $\varphi_N = c_+ \lambda_+^{N-1} + c_- \lambda_-^{N-1}$ for $N \geq 1$, with $c_{\pm}(\theta)$ determined by $\varphi_1 = \cos \theta$ and $\varphi_2 = p \cos 2\theta + q$:

$$c_+ = \frac{\varphi_2 - \lambda_- \cos \theta}{\lambda_+ - \lambda_-}, \quad c_- = \cos \theta - c_+.$$

As $\theta \rightarrow 0$ we have $\lambda_+ \rightarrow 1$, $\lambda_- \rightarrow \rho$, $c_+ \rightarrow 1$, $c_- \rightarrow 0$, and

$$\lambda_+(\theta) = p \left(1 - \frac{\theta^2}{2} \right) + q \left(1 - \frac{p^2 \theta^2}{2q^2} \right) + O(\theta^4) = 1 - \frac{p}{2q} \theta^2 + O(\theta^4).$$

Fix $s \in \mathbb{R}$ and put $\theta = s/\sqrt{N}$. Since $|\rho| < 1$, $c_- \lambda_-^{N-1} \rightarrow 0$, and $\lambda_+^{N-1} \rightarrow \exp(-\frac{p}{2q} s^2)$. Hence, $\varphi_N(s/\sqrt{N}) \rightarrow \exp(-\frac{p}{2q} s^2)$, and by Lévy's continuity theorem [9, Theorem 26.3] $X_N/\sqrt{N}$ converges to the normal law with variance p/q . Finally $K_N - N/2 = X_N/2$. $\square$

For $p = 1$ the ball is in bin 0 or bin N with probability $\frac{1}{2}$ each, and for $p = 0$ it is in a middle bin, so there is no normal limit in these two cases.

Remark 4.3 (cylinder). On a cylinder whose rows have c pegs the local rule is unchanged, so the position is X_N modulo $2c$ (in half-peg units), and the discrete Fourier coefficients of its law are $\varphi_N(\pi l/c)$, $0 \leq l < 2c$. The formula $\varphi_N = c_+ \lambda_+^{N-1} + c_- \lambda_-^{N-1}$ holds whenever $\lambda_+ \neq \lambda_-$. For $0 < p < 1$ and $\theta \notin \pi\mathbb{Z}$ both eigenvalues have modulus less than 1: real eigenvalues satisfy $|\lambda_\pm| \leq p|\cos \theta| + q$, and complex ones have modulus $\sqrt{p}$. The coefficients at $l = 0$ and $l = c$ are 1 and $(-1)^N$. Hence, on the cylinder the law approaches the uniform law on the c positions of the correct parity geometrically fast, and there is no normal shape in the limit.

Remark 4.4 (tetrahedron). For the board with d directions of Proposition 2.3 let K_a be the number of bounces in direction a . The direction chain has transition matrix $\sigma I + (1 - \sigma)\frac{1}{d}J$ with J the all-ones matrix, so the indicator of direction a at time i and of direction b at time $i + m$ have covariance $\sigma^m(\frac{\delta_{ab}}{d} - \frac{1}{d^2})$, and as in Proposition 4.1

$$\text{Cov}(K_a, K_b) = \left(\frac{\delta_{ab}}{d} - \frac{1}{d^2}\right) \left(N \frac{1 + \sigma}{1 - \sigma} - \frac{2\sigma(1 - \sigma^N)}{(1 - \sigma)^2}\right), \quad \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

These formulas hold for $p < 1$; at $p = 1$ the covariance is $N^2(\delta_{ab}/d - 1/d^2)$. The asymptotic multiplier of the multinomial covariance is $(d - 2 + dp)/(d(1 - p))$, which is $(1 + 3p)/(3(1 - p))$ for the tetrahedron.

Theorem 4.5 (simplex normal limit). *Let $d \geq 3$ and $0 \leq p < 1$, or $d = 2$ and $0 < p < 1$. Set*

$$\pi = (1/d, \dots, 1/d), \quad C = \frac{1}{d}I - \frac{1}{d^2}J, \quad c = \frac{1 + \sigma}{1 - \sigma} = \frac{d - 2 + dp}{d(1 - p)}.$$

Then $(K_N - N\pi)/\sqrt{N}$ converges in distribution to $\mathcal{N}_d(0, cC)$. This Gaussian is supported on the sum-zero hyperplane and is nondegenerate there.

Proof. Let $\Pi = \mathbf{1}\pi$ and $T = \sigma I + (1 - \sigma)\Pi$. Fix a real vector a and put $f_i = a_i - d^{-1}\sum_j a_j$, so $\pi f = 0$ and $Tf = \sigma f$. For $D(t) = \text{diag}(e^{itf_i})$ and $B(t) = TD(t)$, stationarity gives

$$\mathbb{E} \exp \left(it \sum_{j=1}^N f(X_j) \right) = \pi D(t) B(t)^{N-1} \mathbf{1}.$$

At zero, $B(0) = T$ has the simple eigenvalue 1 and all other eigenvalues equal σ , with $|\sigma| < 1$. Analytic perturbation gives an eigenvalue $\lambda(t)$ and right eigenvector $h(t)$ with $\lambda(0) = 1$, $h(0) = \mathbf{1}$ and $\pi h(t) = 1$. The complementary spectral part is uniformly exponentially small for sufficiently small t : choose a circle of radius $\eta < 1$ enclosing the complementary spectrum but not $\lambda(t)$ and use the resolvent formula for matrix powers. Thus the characteristic function equals $b(t)\lambda(t)^{N-1} + O(\eta^N)$ with $b(t) \rightarrow 1$.

Differentiate $Bh = \lambda h$ at zero. Since $B'(0) = iT \text{diag}(f)$,

$$\lambda'(0) = 0, \quad (I - T)h'(0) = i\sigma f, \quad h'(0) = \frac{i\sigma}{1 - \sigma} f.$$

Differentiating again and multiplying by π gives

$$\lambda''(0) = \pi B''(0)\mathbf{1} + 2\pi B'(0)h'(0) = -\frac{1 + \sigma}{1 - \sigma} \pi(f^2) = -c\pi(f^2).$$

Hence $\lambda(t) = 1 - c\pi(f^2)t^2/2 + O(t^3)$. Substitute $t = s/\sqrt{N}$ to obtain the scalar normal characteristic function with variance $c\pi(f^2) = ca^\top Ca$. Lévy's theorem and the Cramér–Wold device give the result. $\square$

For $d = 3$ the first two centered coordinates have limiting covariance

$$\frac{1+3p}{27(1-p)} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix};$$

the third is minus their sum. Unlike the binary case, $p = 0$ is included for $d \geq 3$. At $d = 2, p = 0$ bounded alternation gives the zero limit after division by $\sqrt{N}$. At $p = 1$ the centered sequence on this scale is not tight. This theorem asserts weak convergence, not a local approximation for individual bin probabilities.

5 When does the middle equal the extremes?

Let $N \geq 2$ and let $m_N = \lfloor N/2 \rfloor$ be a middle bin (for odd N the two middle bins have equal probability by symmetry). Put $u = q/p$. By Theorem 2.1,

$$\frac{P_N(m_N)}{P_N(0)} = S_N(u) := \sum_{j=1}^{N-1} W(N, m_N, j) u^j, \quad (5)$$

a polynomial in u with nonnegative integer coefficients and $W(N, m_N, 1) = 2$.

Theorem 5.1. (a) For every $N \geq 2$ there is exactly one $p = p_N \in [0, 1]$ with $P_N(m_N) = P_N(0)$.

The middle is more likely than the extremes for $p < p_N$ and less likely for $p > p_N$.

(b) $p_2 = \frac{2}{3}$, $p_3 = 1/\sqrt{2}$, p_4 is the real root of $3p^3 - 4p^2 + 4p - 2$, and $p_N < p_{N+1}$ for all $N \geq 2$.

(c) $p_N \rightarrow 1$, and more precisely $1 - p_N \sim (\log N)/N$.

Proof. (a) At $p = 0$ the extremes have probability 0 and the middle does not. For $p > 0$ the equation is $S_N(u) = 1$ by (5). S_N is continuous and strictly increasing on $[0, \infty)$ with $S_N(0) = 0$ and $S_N(u) \geq 2u$, so there is exactly one root $u_N \in (0, \frac{1}{2}]$, and $p = 1/(1+u)$ is a decreasing bijection from $[0, \infty)$ to $(0, 1]$.

(b) $S_2 = 2u$, $S_3 = 2u + u^2$ and $S_4 = 2u + 2u^2 + 2u^3$ give the three values; for p_4 substitute $u = (1-p)/p$ in $S_4(u) = 1$. For monotonicity we compare coefficients. If $N = 2m$ then the middle $(a, b) = (m, m)$ becomes $(m, m+1)$ for $N+1$, and if $N = 2m+1$ then $(m, m+1)$ becomes $(m+1, m+1)$. In both cases one of a, b grows by one and the other is unchanged, the formulas of Theorem 2.1 are symmetric in (a, b) , and every binomial coefficient $\binom{a-1}{\cdot}$ is nondecreasing in a . So $W(N+1, m_{N+1}, j) \geq W(N, m_N, j)$ for every j . The coefficient of u^2 is $a+b-2 = N-2$, which increases strictly. Hence, $S_{N+1}(u) > S_N(u)$ for $u > 0$, so $u_{N+1} < u_N$.

(c) Let $a = \lfloor N/2 \rfloor$ and $b = \lceil N/2 \rceil$, and fix $\varepsilon \in (0, \frac{1}{2})$.

Lower bound for u_N . Using $\binom{a-1}{r} \leq b^r/r!$ and writing $x = bu$, the terms of $S_N(u)$ with $j = 2i-1$ are at most $\frac{2}{b} x(x^{i-1}/(i-1)!)^2$ and those with $j = 2i$ are at most $\frac{2}{b} \frac{x^i}{i!} \frac{x^i}{(i-1)!}$. Since $\sum_r (x^r/r!)^2 \leq e^{2x}$ and $\sum_i \frac{x^i}{i!} \frac{x^i}{(i-1)!} \leq e^x \cdot x e^x$,

$$S_N(u) \leq \frac{4x}{b} e^{2x}.$$

If $x \leq (\frac{1}{2} - \varepsilon) \log b$ the right side is at most $2b^{-2\varepsilon} \log b$, which is less than 1 for large b . Hence, $b u_N > (\frac{1}{2} - \varepsilon) \log b$ for large N .

Upper bound for u_N . Let $u^* = (\frac{1}{2} + \varepsilon)(\log a)/a$, $y = au^*$ and $r = \lfloor y \rfloor$. Keep only the term $j = 2r + 1$ of $S_N(u^*)$ and use $\binom{b-1}{r} \geq \binom{a-1}{r} \geq (a-r)^r/r!$:

$$S_N(u^*) \geq 2u^* \left(\frac{(a-r)^r (u^*)^r}{r!} \right)^2 = \frac{2y}{a} \left(1 - \frac{r}{a} \right)^{2r} \left(\frac{y^r}{r!} \right)^2.$$

From $\log r! \leq \int_1^r \log t \, dt + \log r$ we get $r! \leq e r (r/e)^r$, so $y^r/r! \geq r^r/r! \geq e^{r-1}/r \geq e^{y-2}/y$. Also $(1 - r/a)^{2r} \geq 1 - 2r^2/a \rightarrow 1$. Therefore

$$S_N(u^*) \geq (1 - o(1)) \frac{2e^{2y-4}}{a y} = (1 - o(1)) \frac{2e^{-4} a^{2\varepsilon}}{y} \rightarrow \infty,$$

so $S_N(u^*) > 1$ and $u_N < u^*$ for large N .

Since $a, b = N/2 + O(1)$, the two bounds give $u_N \sim \log(N/2)/N \sim (\log N)/N$, and $1 - p_N = u_N/(1 + u_N) \sim u_N$. $\square$

In Kagey's terms: for the $(2n+1)$ -st row ($N = 2n$) the unique value is p_{2n} , and the least of these is $p_2 = \frac{2}{3}$, which is why the third row of his $p = \frac{2}{3}$ table is 2, 2, 2. For the $(2n)$ -th row ($N = 2n - 1$, $n \geq 2$) the least is $p_3 = 1/\sqrt{2}$; the second row is 1, 1 for every p . There is no greatest value, since $p_N \uparrow 1$. The tenth row has $p_9 = 0.82829\dots$, just below $\frac{5}{6}$, which is why the extremes and the middle are nearly level in his third histogram.

N	p_N	N	p_N	N	p_N	N	p_N
2	0.66666667	6	0.78871211	10	0.83840411	50	0.94200405
3	0.70710678	7	0.80361731	11	0.84652725	100	0.96496633
4	0.74487450	8	0.81759938	12	0.85426060	200	0.97939018
5	0.76759188	9	0.82829163	20	0.89293837	400	0.98812464

Table 1: The value p_N at which the middle and extreme bins of row $N + 1$ are equally likely.

Theorem 5.2 (uniform Bessel approximation). *Put $h = N/2$, $J(x) = I_0(2x) + I_1(2x)$ and $F(x) = 2xJ(x)$. For every $N \geq 2$ and $x > 0$,*

$$0 \leq 1 - \frac{hS_N(x/h)}{F(x)} \leq \frac{x(x+1)}{h}. \quad (6)$$

Let x_B be the unique positive root of $F(x_B) = h$, and put $u_B = x_B/h$, $u_N = (1 - p_N)/p_N$. Then

$$0 \leq u_N - u_B = O((\log N)^2/N^2).$$

Writing $L = \log N$ and $c_0 = \frac{1}{2} \log(\pi/8)$, we have

$$N(1 - p_N) = L - \frac{1}{2} \log L + c_0 + O\left(\frac{\log L}{L}\right), \quad (7)$$

$$N(1 - p_N) = L - \frac{1}{2} \log L + c_0 + \frac{\frac{1}{4} \log L - \frac{1}{2} c_0 + \frac{1}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right). \quad (8)$$

Proof. Let $a = \lfloor N/2 \rfloor$, $b = \lceil N/2 \rceil$, and define

$$A_r = \frac{r!}{h^r} \binom{a-1}{r}, \quad B_r = \frac{r!}{h^r} \binom{b-1}{r} \quad (r \geq 0).$$

With $(v)_+ = \max(v, 0)$, these have the product expressions

$$A_r = \prod_{k=1}^r \left(1 - \frac{h-a+k}{h}\right)_+, \quad B_r = \prod_{k=1}^r \left(1 - \frac{h-b+k}{h}\right)_+.$$

They lie in $[0, 1]$, including beyond their finite support, where a factor is zero. For nonnegative t_j , the inequality $1 - \prod_j (1 - t_j)_+ \leq \sum_j t_j$ holds even when some $t_j > 1$. Since $a + b = 2h$, it gives

$$0 \leq 1 - A_r B_r \leq \frac{r(r+1)}{h}.$$

The deficits for $A_{r+1}B_r$ and $A_r B_{r+1}$ are at most $((r+1)^2 + h - a)/h$ and $((r+1)^2 + h - b)/h$, respectively. Averaging cancels the parity offsets, so

$$0 \leq 1 - \frac{A_{r+1}B_r + A_r B_{r+1}}{2} \leq \frac{(r+1)^2}{h}.$$

The run-count formula, with $u = x/h$, becomes

$$S_N(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} A_r B_r + \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} \frac{A_{r+1}B_r + A_r B_{r+1}}{2} \right].$$

Dropping the A, B factors yields $2uJ(x) = F(x)/h$, using the Bessel series [20, Eq. 10.25.2]. Index shifts in those absolutely convergent series give

$$\begin{aligned} \sum_{r \geq 0} r(r+1) \frac{x^{2r}}{(r!)^2} &= x^2 I_0(2x) + x I_1(2x), \\ \sum_{r \geq 0} (r+1)^2 \frac{x^{2r+1}}{r!(r+1)!} &= x^2 I_1(2x) + x I_0(2x). \end{aligned}$$

Their sum is $x(x+1)J(x)$, proving (6), including the tail beyond the finite polynomial support. In particular the relative error is $O((\log N)^2/N)$ uniformly for $0 < x \leq C \log N$ for each fixed C .

Now let $x_N = hu_N$. Theorem 5.1 gives $x_N \sim \frac{1}{2} \log N$. Differentiation gives $F'(x) = 2F(x) + 2I_0(2x)$, hence $F'/F \geq 2$. Set $\delta_N = x_N(x_N + 1)/h$. For large N , $\delta_N < 1$, and (6) at the root gives

$$h \leq F(x_N) \leq \frac{h}{1 - \delta_N}, \quad 0 \leq x_N - x_B \leq -\frac{1}{2} \log(1 - \delta_N) = O((\log N)^2/N).$$

Divide by h for the assertion about u . Moreover,

$$N(1 - p_N) = \frac{2x_N}{1 + x_N/h} = 2x_B + O((\log N)^2/N).$$

Put $z = 2x_B$. The standard large-argument expansion [20, Eq. 10.40.1] is

$$I_0(z) + I_1(z) = \frac{2e^z}{\sqrt{2\pi z}} \left(1 - \frac{1}{8z} + O(z^{-2})\right).$$

Since $z(I_0(z) + I_1(z)) = N/2$, taking logarithms gives

$$z + \frac{1}{2} \log z = L + c_0 + \frac{1}{8z} + O(z^{-2}).$$

First $z = L + O(\log L)$, then substitution proves (7). For one further term put $z = L + d + e/L$, where $d = -\frac{1}{2} \log L + c_0$. Matching coefficients of $1/L$ gives $e = \frac{1}{8} - d/2$. The residual is $O((\log L)^2/L^2)$; the derivative of $z + \frac{1}{2} \log z$ is bounded below by one, so inversion preserves that order. This proves (8). $\square$

The ratios u_B/u_N are 0.9761, 0.9864, 0.9924 at $N = 100, 200, 400$ and 0.99579, 0.99769, 0.99874 at $N = 800, 1600, 3200$. The latter three were predicted within registered intervals before computation (`data/ledger.md`). Theorem 5.2 now supplies the proof previously missing from that heuristic.

Proposition 5.3 (adjacent bins and global extrema). *For $N \geq 2$, an adjacent bin has the same probability as an endpoint precisely at*

$$p_N^{\text{adj}} = \frac{1 + \sqrt{N-1}}{2 + \sqrt{N-1}}.$$

For $0 < p < 1$, endpoints attain the global minimum if $p < p_N^{\text{adj}}$; bins $1, N-1$ attain it if $p > p_N^{\text{adj}}$; both sets do at equality. At $p = p_N$ the middle and endpoints attain the global maximum and bins $1, N-1$ attain the global minimum, allowing additional ties in small rows.

Proof. A word with one R has one change in two cases and two changes in $N-2$ cases. Thus $P_N(1)/P_N(0) = 2u + (N-2)u^2$, and its positive root at level one is $u = 1/(1 + \sqrt{N-1})$, including $N=2$. This yields the stated p and sign change. Dekking and Kong [10, Proposition 4.3] prove log-concavity on the interior bins. Symmetry therefore makes the middle maximal and bins $1, N-1$ minimal among interior bins. Comparing those minima with the endpoints proves the claims. $\square$

The adjacent-bin scale is $1 - p_N^{\text{adj}} \sim N^{-1/2}$, distinct from $1 - p_N \sim (\log N)/N$. At $p = 0, 1$ additional zero-probability ties occur.

6 The full hierarchy of endpoint crossings

The middle and adjacent bins are the two ends of a larger family of crossing thresholds. In this section, a, b are positive integers, $N = a + b$, and

$$S_{a,b}(u) = \frac{P_{a+b}(a)}{P_{a+b}(0)}, \quad u = \frac{1-p}{p}.$$

This ratio is defined for $p > 0$. The run-count formula gives the useful form

$$S_{a,b}(u) = \sum_{r \geq 0} \binom{a-1}{r} \binom{b-1}{r} \left(2u^{2r+1} + \frac{N-2-2r}{r+1} u^{2r+2} \right). \quad (9)$$

Indeed, the even coefficient follows from

$$\binom{a-1}{r+1} \binom{b-1}{r} + \binom{a-1}{r} \binom{b-1}{r+1} = \frac{N-2-2r}{r+1} \binom{a-1}{r} \binom{b-1}{r}.$$

The summation is finite and all its coefficients are nonnegative. Since the linear coefficient is 2, there is a unique $u_{a,b} > 0$ satisfying $S_{a,b}(u_{a,b}) = 1$. Write

$$p_{a,b} = \frac{1}{1 + u_{a,b}}.$$

Thus every interior bin has exactly one endpoint crossing in $0 < p < 1$. The qualification is necessary: at $p = 0$, a noncentral bin and an endpoint can both have probability zero.

Proposition 6.1 (ordering the thresholds). *If $a < b - 1$, then*

$$u_{a+1,b-1} < u_{a,b}, \quad p_{a+1,b-1} > p_{a,b}.$$

Also $p_{a,b+1} > p_{a,b}$ for all positive a, b . In fact, the corresponding inequalities between the polynomials $S_{a+1,b-1}$ and $S_{a,b}$, or $S_{a,b+1}$ and $S_{a,b}$, hold coefficientwise, with at least one strict coefficient.

Proof. For fixed N and $r \leq a - 1$, write

$$\binom{a-1}{r} \binom{b-1}{r} = \frac{1}{(r!)^2} \prod_{j=0}^{r-1} (a-1-j)(b-1-j).$$

Every factor increases after (a, b) is replaced by $(a+1, b-1)$, since

$$(a-j)(b-2-j) - (a-1-j)(b-1-j) = b-a-1 > 0.$$

Any new summand is nonnegative, and the cubic coefficient increases strictly. Equation (9) therefore gives the first assertion. Increasing b with a fixed increases the binomial coefficients in the original run formula, and increases the quadratic coefficient $N-2$ strictly. Comparing the unique roots proves the remaining assertions. $\square$

The ordering of the bin heights is already a consequence of the interior log-concavity proved by Dekking and Kong [10, Proposition 4.3]. The coefficient argument is included as an elementary strengthening and as a convenient tool below. The quantitative statements that follow address the locations of all the crossings, including bins whose distance from the edge changes with N .

6.1 A Bessel estimate uniform in the bin

Put

$$s = \sqrt{ab}, \quad A = \frac{ab}{a+b}, \quad c = \frac{a+b}{2\sqrt{ab}}, \quad \eta = \frac{1}{c} = \frac{2\sqrt{ab}}{a+b}, \quad x = su.$$

Define

$$J_c(x) = I_0(2x) + cI_1(2x), \quad T_{a,b}(u) = 2uJ_c(su).$$

Theorem 6.2 (uniform estimate for every bin). *For all positive integers a, b and all $u > 0$,*

$$0 \leq 1 - \frac{S_{a,b}(u)}{T_{a,b}(u)} \leq \frac{(a+b)u^2}{2} + u = \frac{x^2}{2A} + \frac{x}{s}. \quad (10)$$

In particular, at fixed (N, u) the relative-error bound is independent of the bin index. The inequality remains valid when its right side exceeds one, although it then gives no positive lower bound for $S_{a,b}$.

Proof. Set

$$U_r = \frac{r!}{a^r} \binom{a-1}{r}, \quad V_r = \frac{r!}{b^r} \binom{b-1}{r}, \quad H = \frac{1}{a} + \frac{1}{b}.$$

These numbers lie in $[0, 1]$. Their product representations, using positive parts beyond the finite support, and $1 - \prod_i (1 - t_i)_+ \leq \sum_i t_i$ for $t_i \geq 0$, give

$$0 \leq 1 - U_r V_r \leq \frac{H}{2} r(r+1). \quad (11)$$

For the even terms define

$$E_r = \frac{aU_{r+1}V_r + bU_rV_{r+1}}{a+b}.$$

The same argument, with the indicated weights, gives

$$\begin{aligned} 0 \leq 1 - E_r &\leq \frac{(r+1)(r+2)}{a+b} + \frac{r(r+1)}{2(a+b)} \left(\frac{a}{b} + \frac{b}{a} \right) \\ &= \frac{H}{2}r(r+1) + \frac{2(r+1)}{a+b}. \end{aligned} \quad (12)$$

The exact polynomial is

$$S_{a,b}(u) = 2u \left[\sum_{r \geq 0} \frac{x^{2r}}{(r!)^2} U_r V_r + c \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} E_r \right]. \quad (13)$$

Dropping the product factors gives $T_{a,b}$. Index shifts in the Bessel series yield

$$\begin{aligned} \sum_{r \geq 0} r(r+1) \frac{x^{2r}}{(r!)^2} &= x^2 I_0(2x) + x I_1(2x), \\ \sum_{r \geq 0} r(r+1) \frac{x^{2r+1}}{r!(r+1)!} &= x^2 I_1(2x), \\ \sum_{r \geq 0} (r+1) \frac{x^{2r+1}}{r!(r+1)!} &= x I_0(2x). \end{aligned}$$

Consequently the error in (13), after division by $2u$, is at most

$$\frac{H}{2} x^2 J_c(x) + x \left\{ \frac{H}{2} I_1(2x) + \frac{2c}{a+b} I_0(2x) \right\} = \left(\frac{x^2}{2A} + \frac{x}{s} \right) J_c(x).$$

Here $H/2 = c/s = 1/(2A)$ and $2c/(a+b) = 1/s$. This proves the result, including the infinite tail beyond the polynomial's finite support. $\square$

Let $x_{a,b} = su_{a,b}$, and let x_B be the unique positive root of

$$G_\eta(x_B) = A, \quad G_\eta(x) = x\{I_1(2x) + \eta I_0(2x)\}. \quad (14)$$

The positive series makes G_η strictly increasing. Standard Bessel identities also give

$$G'_\eta(x) - 2G_\eta(x) = 2x(1-\eta)\{I_0(2x) - I_1(2x)\} + \eta I_0(2x) \geq 0. \quad (15)$$

The sign $I_0(2x) > I_1(2x) > 0$ follows directly from their integral representations. Thus the surrogate has a uniformly controlled inverse.

Corollary 6.3 (quantitative crossing bound). *We have $x_{a,b} \geq x_B$. If*

$$\delta_{a,b} = \frac{x_{a,b}^2}{2A} + \frac{x_{a,b}}{s} < 1,$$

then

$$0 \leq x_{a,b} - x_B \leq -\frac{1}{2} \log(1 - \delta_{a,b}). \quad (16)$$

Proof. At the exact crossing, Theorem 6.2 gives $A \leq G_\eta(x_{a,b}) \leq A/(1 - \delta_{a,b})$. Integrate $(\log G_\eta)' \geq 2$, which follows from (15). $\square$

6.2 Every growing distance from an edge

Theorem 6.4 (uniform crossing expansion). *Suppose $1 \leq a \leq b$ and $a \rightarrow \infty$, with no restriction on the relative growth of a and b . Put*

$$L = \log A, \quad C(\eta) = \frac{1}{2} \log(8\pi) - \log(1 + \eta), \quad B(\eta) = \frac{3 - \eta}{8(1 + \eta)}.$$

Uniformly over $b \geq a$,

$$\begin{aligned} 2\sqrt{ab}u_{a,b} &= L - \frac{1}{2} \log L + C(\eta) \\ &\quad + \frac{\frac{1}{4} \log L - \frac{1}{2} C(\eta) + B(\eta)}{L} + O\left(\frac{(\log L)^2}{L^2} + \frac{(\log a)^2}{a}\right). \end{aligned} \quad (17)$$

The same expansion and error order hold with $1 - p_{a,b}$ in place of $u_{a,b}$.

Proof. We have $a/2 \leq A \leq a$, $s \geq a$, and $0 < \eta \leq 1$. At $x = \log a$, the relative error in Theorem 6.2 tends to zero uniformly, whereas

$$\frac{G_\eta(x)}{A} \geq \frac{xI_1(2x)}{a} \rightarrow \infty.$$

The last assertion follows from the standard positive-real Bessel expansion. Hence $x_{a,b} \leq \log a$ for all sufficiently large a , uniformly over b . Corollary 6.3 now implies

$$x_{a,b} - x_B = O((\log a)^2/a). \quad (18)$$

Write $z = 2x_B$. The usual large-argument expansions [20, Eq. 10.40.1], combined with $0 \leq \eta \leq 1$, give uniformly

$$I_1(z) + \eta I_0(z) = \frac{(1 + \eta)e^z}{\sqrt{2\pi z}} \left\{ 1 + \frac{\eta - 3}{8(1 + \eta)z} + O(z^{-2}) \right\}.$$

Thus (14) becomes

$$z + \frac{1}{2} \log z = L + C(\eta) + \frac{B(\eta)}{z} + O(z^{-2}).$$

Initially this gives $z = L + O(\log L)$ uniformly. Substitution of $z = L + d + e/L$ then gives

$$d = -\frac{1}{2} \log L + C(\eta), \quad e = \frac{1}{4} \log L - \frac{1}{2} C(\eta) + B(\eta).$$

The residual is $O((\log L)^2/L^2)$; the derivative of $z + \frac{1}{2} \log z$ is at least one. Combining this inversion with (18) proves (17). Finally,

$$2s \left(u - \frac{u}{1 + u} \right) = O(su^2) = O((\log a)^2/s) = O((\log a)^2/a),$$

which proves the assertion about $1 - p_{a,b}$. □

In particular,

$$1 - p_{a,b} \sim \frac{\log a}{2\sqrt{ab}} \quad (a \rightarrow \infty, b \geq a), \quad (19)$$

uniformly in b . If $a = o(b)$, the more precise form is

$$2\sqrt{ab}(1 - p_{a,b}) = \log a - \frac{1}{2} \log \log a + \frac{1}{2} \log(8\pi) + o(1). \quad (20)$$

For a fixed bulk fraction $a/N \rightarrow \alpha \in (0, 1/2]$, the corresponding formula is

$$2\sqrt{a(N-a)}(1-p_{a,N-a}) = \log N - \frac{1}{2} \log \log N + \log\{\alpha(1-\alpha)\} \\ + \frac{1}{2} \log(8\pi) - \log\{1 + 2\sqrt{\alpha(1-\alpha)}\} + o(1).$$

The exact prefactor is retained because no convergence rate for a/N was assumed. At the center this recovers the constant $\frac{1}{2} \log(\pi/8)$. There is no separate requirement such as $a \gg \log^2 N$: the relevant Bessel argument at an edge is of order $\log a$, and its depletion error is $O(\log^2(a)/a)$ for every growing a .

6.3 Fixed bins and the Laguerre edge law

For a positive integer a , define

$$H_a(y) = \sum_{r=0}^{a-1} \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = \frac{y}{a} L_{a-1}^{(1)}(-y), \quad (21)$$

where the second expression uses the associated Laguerre polynomial with the convention specified by the finite sum. Let y_a be the unique positive root of $H_a(y_a) = 1$, and put $c_a = \sqrt{y_a}$. We have $y_1 = 1$, $y_2 = \sqrt{3} - 1$, and y_a decreases strictly with a , by coefficient comparison.

Theorem 6.5 (fixed-edge expansion). *For each fixed positive integer a , as $b \rightarrow \infty$,*

$$u_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1}{b} + \frac{\gamma_a}{b^{3/2}} + O_a(b^{-2}), \quad (22)$$

where

$$\gamma_a = \frac{1 + y_a + (y_a + y_a^2/2)H_a''(y_a)/H_a'(y_a)}{2c_a}.$$

Thus the second coefficient in the odds expansion is -1 for every fixed bin. For the switching probability,

$$1 - p_{a,b} = \frac{c_a}{\sqrt{b}} - \frac{1 + c_a^2}{b} + O_a(b^{-3/2}). \quad (23)$$

Proof. Put $\epsilon = b^{-1/2}$, $y = bu^2$, and $V_r = \prod_{j=1}^r (1 - j/b)$. With a fixed and $b \geq a$, the exact run polynomial can be written

$$S_{a,b}(u) = 2\sqrt{y/b} \sum_r \binom{a-1}{r} \frac{y^r}{r!} V_r \\ + \frac{y}{b} \sum_r \binom{a-1}{r+1} \frac{y^r}{r!} V_r + y \sum_r \binom{a-1}{r} \frac{y^r}{(r+1)!} V_{r+1}. \quad (24)$$

All these sums have bounded length. Expanding $V_r = 1 - r(r+1)/(2b) + O_a(b^{-2})$ gives, uniformly near $y = y_a$,

$$S_{a,b}(u) = H_a(y) + 2\sqrt{y} H_a'(y) \epsilon \\ + \{(y - y^2/2)H_a''(y) - yH_a'(y)\}\epsilon^2 + O_a(\epsilon^3). \quad (25)$$

In particular, the second-order term uses the identity

$$\sum_r (r+1)(r+2) \binom{a-1}{r} \frac{y^{r+1}}{(r+1)!} = y^2 H_a''(y) + 2y H_a'(y).$$

The finite-product expression (24) extends analytically in (ϵ, y) near $(0, y_a)$, and $H_a'(y_a) > 0$. The implicit-function theorem therefore gives the unique crossing as

$$y = y_a + d_1 \epsilon + d_2 \epsilon^2 + O_a(\epsilon^3).$$

Matching coefficients in (25) gives

$$d_1 = -2c_a, \quad d_2 = 2 + y_a + (y_a + y_a^2/2) \frac{H_a''(y_a)}{H_a'(y_a)}.$$

Expanding $u = \epsilon \sqrt{y}$ proves (22). Expanding $u/(1+u)$ gives (23). $\square$

The edge law matches the growing-bin law quantitatively; the two limits do not require an unproved interchange.

Theorem 6.6 (matching the edge and Bessel regimes). *For every positive integer a and every $x > 0$,*

$$0 \leq 1 - \frac{a H_a(x^2/a)}{x I_1(2x)} \leq \frac{x^2}{2a}. \quad (26)$$

Consequently, with $\ell = \log a$ and $C_0 = \frac{1}{2} \log(8\pi)$,

$$\begin{aligned} 2\sqrt{ay_a} &= \ell - \frac{1}{2} \log \ell + C_0 \\ &\quad + \frac{\frac{1}{4} \log \ell - \frac{1}{2} C_0 + 3/8}{\ell} + O\left(\frac{(\log \ell)^2}{\ell^2} + \frac{(\log a)^2}{a}\right). \end{aligned} \quad (27)$$

Proof. With $U_r = (r!/a^r) \binom{a-1}{r}$ as before,

$$H_a(x^2/a) = \frac{x}{a} \sum_{r \geq 0} \frac{x^{2r+1}}{r!(r+1)!} U_r.$$

Apply $0 \leq 1 - U_r \leq r(r+1)/(2a)$ and the identity

$$\sum_{r \geq 0} r(r+1) \frac{x^{2r+1}}{r!(r+1)!} = x^2 I_1(2x)$$

to obtain (26). The same root comparison as in Corollary 6.3 compares $\sqrt{ay_a}$ with the root of $x I_1(2x) = a$, with error $O(\log^2(a)/a)$. Invert that equation as in Theorem 6.4, now with $\eta = 0$ and $A = a$. Here $C(0) = C_0$ and $B(0) = 3/8$, which proves the expansion. $\square$

6.4 The moving boundary layer

The all-bin threshold theorem also describes the shape of a row at a prescribed switching probability. Let $v_N > 0$ be the chosen odds, so $p = 1/(1 + v_N)$, and define

$$\mathcal{L}_N = \#\{1 \leq k \leq \lfloor N/2 \rfloor : P_N(k) < P_N(0)\}.$$

By Proposition 6.1, these bins form an initial interval adjacent to the endpoint.

Corollary 6.7 (boundary-layer width). *Suppose*

$$T_N = \frac{1}{\sqrt{N} v_N} \rightarrow \infty, \quad \frac{\log T_N}{N v_N} \rightarrow 0.$$

Then

$$\mathcal{L}_N \sim T_N^2 \log^2 T_N = \frac{\log^2(1/(\sqrt{N} v_N))}{N v_N^2}. \quad (28)$$

In particular, if $v_N = N^{-\gamma}$ with $1/2 < \gamma < 1$, then

$$\mathcal{L}_N \sim (\gamma - \tfrac{1}{2})^2 N^{2\gamma-1} \log^2 N.$$

By symmetry, the total number of interior bins below the endpoint height is $2\mathcal{L}_N$ for all sufficiently large N .

Proof. Set $d_N = T_N^2 \log^2 T_N$. The assumptions give $d_N \rightarrow \infty$ and

$$\frac{d_N}{N} = \left(\frac{\log T_N}{N v_N} \right)^2 \rightarrow 0.$$

For each fixed $\varepsilon \in (0, 1)$, choose integers $a_{\pm} = (1 \pm \varepsilon)d_N + O(1)$. Theorem 6.4, or its leading form (19), gives

$$\frac{u_{a_{\pm}, N-a_{\pm}}}{v_N} \sim \frac{\log a_{\pm}}{2\sqrt{a_{\pm}N} v_N} \rightarrow \frac{1}{\sqrt{1 \pm \varepsilon}}.$$

Here $\log a_{\pm} \sim 2 \log T_N$ was used. Thus bin a_- lies below the endpoint height, while bin a_+ lies above it, for all sufficiently large N . Strict ordering then bounds $\mathcal{L}_N$ between these indices. Letting $\varepsilon \downarrow 0$ proves the result. Finally $\mathcal{L}_N = o(N)$, so the reflected intervals are disjoint and contain every bin below the endpoints. $\square$

The results in this section use classical run enumeration and Bessel asymptotics. Their potential novelty lies in the explicit uniform error, the complete range of crossing thresholds, and the matched boundary laws; the literature qualifications in the introduction continue to apply.

7 Boundary conditions and a spatial crossing window

The binary model is also a zero-field Ising chain with free boundaries: for step signs $\varepsilon_i \in \{-1, 1\}$ its weight is proportional to $\exp(K \sum_{i=1}^{N-1} \varepsilon_i \varepsilon_{i+1})$, where $u = q/p = e^{-2K}$. The analogous periodic measure includes the last-to-first bond. Antal, Droz and Rácz [14] derive boundary-dependent limiting magnetization laws; Dantchev and Rudnick [15] give exact free and periodic coefficients. Stepanyan et al. [16] explicitly study the periodic endpoint-adjacent and endpoint-center comparisons. Thus neither the periodic law nor the crossing question is new. Our purpose is to give a uniform discrete estimate and a rigorous large- N refinement of the reported finite-size approximation.

The periodic spin measure differs from wrapping the spatial position of the free Galton walk around a cylinder; the latter remains the pushforward in Remark 4.3. Write $R_{N,k}^F$ and $R_{N,k}^P$ for bin-to-endpoint ratios under free and periodic boundaries. Put $a = k$, $b = N - k$ and $s = \sqrt{ab}$. The classical cyclic count is

$$R_{N,k}^P(u) = \sum_{r=1}^{\min(a,b)} \frac{N}{r} \binom{a-1}{r-1} \binom{b-1}{r-1} u^{2r}. \quad (29)$$

Indeed, choose compositions of each letter count into r positive runs, choose a labeled position for the start of a distinguished run of one letter, and divide by its r possible choices. The normalization of the periodic measure cancels in the ratio. In particular $R_{N,1}^P = Nu^2$, giving the known adjacent threshold $K = (\log N)/4$. Every interior ratio is strictly increasing from zero to infinity in $u > 0$, and the ratios increase strictly toward the middle, except for the symmetric middle-bin tie. The coefficient comparison uses the same products $(a-j)(b-j)$ as the free case.

Theorem 7.1 (periodic relative error). *For $N \geq 2$, $1 \leq k \leq N-1$, and $u > 0$, define*

$$T_{N,k}^P(u) = \frac{Nu}{\sqrt{k(N-k)}} I_1\left(2u\sqrt{k(N-k)}\right).$$

Then

$$0 \leq 1 - \frac{R_{N,k}^P(u)}{T_{N,k}^P(u)} \leq \frac{Nu^2}{2}. \quad (30)$$

Proof. Set $x = su$, $A = ab/N$, and $U_r = r! \binom{a-1}{r} / a^r$, $V_r = r! \binom{b-1}{r} / b^r$. The clipped-product inequality gives $0 \leq 1 - U_r V_r \leq r(r+1)/(2A)$, including the zero factors beyond support. After shifting the index in (29), the ratio to T^P is the weighted average of $U_r V_r$ with weights $x^{2r+1}/(r!(r+1)!)$. Their sum is $I_1(2x)$, while their $r(r+1)$ -weighted sum is $x^2 I_1(2x)$. Thus the relative deficit is at most $x^2/(2A) = Nu^2/2$. $\square$

Theorem 7.2 (periodic crossing and boundary shift). *Let u_N^P be the positive central root, and let $u_N^F = u_N$ be the free central root. Put $L = \log N$ and $c_P = \frac{1}{2} \log(\pi/2)$. Then*

$$Nu_N^P = L - \frac{1}{2} \log L + c_P + \frac{\frac{1}{4} \log L - \frac{1}{2} c_P + \frac{3}{8}}{L} + O\left(\frac{(\log L)^2}{L^2}\right), \quad (31)$$

$$N(u_N^P - u_N^F) = \log 2 + \frac{\frac{1}{4} - \frac{1}{2} \log 2}{\log N} + o(1/\log N). \quad (32)$$

Replacing Nu_N^P by $N(1 - p_N^P)$ preserves (31). In the Ising convention $K = \beta J$,

$$\beta_N^P J = \frac{1}{2} (\log N - \log \log N) + \frac{\frac{1}{4} \log \log N - \frac{1}{2} c_P}{\log N} + O\left(\frac{(\log \log N)^2}{(\log N)^2}\right). \quad (33)$$

Proof. At the middle, $2s = N + O(1/N)$. Evaluate Theorem 7.1 at $u = c \log N/N$ for fixed $c > 0$. Its relative error tends to zero, and the Bessel expansion makes the ratio tend to zero when $c < 1$ and to infinity when $c > 1$. Monotonicity therefore gives $Nu_N^P \sim \log N$. Writing $z = Nu_N^P$ and using $I_1(z) = e^z (2\pi z)^{-1/2} (1 - 3/(8z) + O(z^{-2}))$ in the estimate at the root yields

$$z + \frac{1}{2} \log z = L + c_P + \frac{3}{8z} + O(z^{-2} + L^2/N).$$

The odd- N argument change contributes only $O(L/N^2)$ to this logarithmic equation. Substitute $z = L + d + e/L$: the constant terms give $d = -\frac{1}{2} \log L + c_P$, and the $1/L$ terms give $e = 3/8 - d/2$. The derivative of the left side is at least one, so the residual yields the stated error. Since $N(u - u/(1+u)) = O(L^2/N)$, the switching-probability version follows. Subtract the free expansion, where $c_F = \frac{1}{2} \log(\pi/8)$ and the correction is $1/8$, to obtain (32). Finally $\beta J = -\frac{1}{2} \log u = \frac{1}{2} (L - \log z)$ gives (33). $\square$

The approximately linear formula in $\log N$ in [16, Eq. (16)] is described there as a finite-size approximation. Equation (33) supplies a rigorous large- N refinement, rather than contradicting a claimed asymptotic theorem.

Theorem 7.3 (spatial crossing window). *Let $X \in \{F, P\}$ and let u_N^X be its exact central crossing. For fixed real t, y set*

$$u = u_N^X + \frac{t}{N}, \quad k_N(y) = \text{nint}\left(\frac{N}{2} + \frac{yN}{\sqrt{\log N}}\right),$$

where either nearest-integer convention may be used. Then, uniformly on compact sets of (t, y) ,

$$R_{N, k_N(y)}^X(u_N^X + t/N) \longrightarrow e^{t-2y^2}. \quad (34)$$

If $M_N^X(t)$ is the number of interior bins with probability strictly above the endpoint probability at that parameter, then

$$\frac{\sqrt{\log N}}{N} M_N^X(t) \longrightarrow \sqrt{2 \max(t, 0)}. \quad (35)$$

Proof. Put $L = \log N$, $\delta = k/N - 1/2$, and $\eta = 2\sqrt{k(N-k)}/N = \sqrt{1-4\delta^2}$. For the stated rounding and bounded y ,

$$\eta = 1 - \frac{2y^2}{L} + O\left(L^{-2} + \frac{1}{N\sqrt{L}}\right), \quad Nu_N^X = L + O(\log L).$$

Thus the Bessel argument at (u, k) equals $(Nu_N^X + t)\eta = Nu_N^X + t - 2y^2 + o(1)$, uniformly on the compact set. Use the free all-bin estimate and Theorem 7.1. Both have relative discrete error $O(L^2/N)$ here. The Bessel expansion $I_\nu(z) = e^z/\sqrt{2\pi z}(1 + O(1/z))$, for $\nu = 0, 1$, shows that the prefactors relative to their central values tend uniformly to one. The argument change contributes e^{t-2y^2} . Since the central exact ratio is one, this proves (34).

The interior ratios increase toward the middle. If $t \leq 0$, no interior bin is strictly above the endpoint at or below the central crossing. If $t > 0$, for any $0 < \epsilon < \sqrt{t/2}$ the bins at scaled distance less than $\sqrt{t/2} - \epsilon$ are above the endpoint, whereas those at distance $\sqrt{t/2} + \epsilon$ are below. Monotonicity excludes all more distant bins, even outside the compact interval on which the local limit was applied. Count the resulting interval, divide by $N/\sqrt{L}$, and let $\epsilon \downarrow 0$. Rounding changes the count by $O(1)$. $\square$

The parameter window has width N^{-1} and the spatial window has width $N/\sqrt{\log N}$. A weak limiting distribution alone does not give this comparison of an individual interior bin with an endpoint atom.

8 Face-center thresholds on a simplex

We retain the symmetric d -direction model of Proposition 2.3: the initial direction is uniform, a repeat has probability p , and each specified change has probability $r = (1-p)/(d-1)$. This specification matters; the results below do not cover every physical interpretation of a tetrahedral board. For $2 \leq k \leq d$ and $N \geq k$, let $C_{N,k}(p)$ denote the probability of one *balanced face bin*: its k positive coordinates differ by at most one, and its other coordinates vanish. All such bins have the same probability. Let $V_N(p) = p^{N-1}/d$ be the probability of one vertex bin, and put

$$u = \frac{1-p}{(d-1)p}, \quad a_k = \frac{1}{2} \log(4\pi) - \frac{k+1/2}{k-1} \log k.$$

The ratio $C_{N,k}/V_N$, viewed as a function of u , depends on k and N but not on the ambient number d of directions.

Theorem 8.1 (face-center crossings). *For every $2 \leq k \leq d$ and $N \geq k$, there is exactly one $p = p_{N,k}^{(d)} \in (0, 1)$ for which $C_{N,k}(p) = V_N(p)$. The face center is more likely below this persistence and less likely above it. The sequence $p_{N,k}^{(d)}$ is strictly increasing in N , and*

$$p_{k,k}^{(d)} = \frac{1}{1 + (d-1)(k!)^{-1/(k-1)}}.$$

Writing $L = \log N$ and $u_{N,k} = (1 - p_{N,k}^{(d)})/((d-1)p_{N,k}^{(d)})$, for fixed k we have

$$Nu_{N,k} = L - \frac{1}{2} \log L + a_k + \frac{\frac{1}{4} \log L - \frac{1}{2} a_k + \frac{1}{8}}{L} + O_k\left(\frac{(\log L)^2}{L^2}\right). \quad (36)$$

Consequently, $N(1 - p_{N,k}^{(d)})$ equals $(d-1)$ times the right-hand side of (36), with the same error order for fixed d .

In particular, for the full center $k = d$,

$$N(1 - p_{N,d}^{(d)}) = (d-1) \log N - \frac{d-1}{2} \log \log N + c_d + o(1), \quad c_d = \frac{d-1}{2} \log(4\pi) - \frac{2d+1}{2} \log d.$$

For $d = 2$, $a_2 = \frac{1}{2} \log(\pi/8)$, and all the displayed coefficients agree with Theorem 5.2. The new feature is that the same moving regime controls every fixed face dimension.

Theorem 8.2 (common critical window). *Fix d and put*

$$u_N(t) = \frac{\log N - \frac{1}{2} \log \log N + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)}.$$

Uniformly for t in any fixed compact interval, for each $2 \leq k \leq d$,

$$\frac{C_{N,k}(p_N(t))}{V_N(p_N(t))} \longrightarrow \exp((k-1)(t - a_k)). \quad (37)$$

Moreover $a_2 > a_3 > \dots$. Thus the face centers cross the vertex height at distinct constant offsets within a common window of persistence width $O(1/N)$.

We first prove the exact crossing assertions and then establish a uniform local estimate that proves both asymptotic theorems. No fixed- p central limit theorem is used to infer probabilities in this moving regime.

Proof of the exact assertions in Theorem 8.1. For a positive count vector $n = (n_1, \dots, n_k)$, let $W_n(j)$ be the number of words with those counts and exactly j changes. Its ratio to a vertex probability is

$$S_n(u) = \sum_j W_n(j) u^j.$$

To count $W_n(j)$, choose a proper word of $j+1$ run colors, meaning that successive colors differ. If color i appears m_i times, its run lengths can be chosen in $\binom{n_i-1}{m_i-1}$ ways. Thus $W_n(j)$ is a sum of products of these binomial coefficients. Every coefficient is nonnegative; the first nonzero one

is $W_n(k-1) = k!$. Hence S_n is strictly increasing from zero to infinity on $u \geq 0$ and crosses one exactly once. At $p = 0$, cycling through the k colors and truncating after N positions gives a word with balanced counts and no successive repeat, so its probability is positive while the vertex probability is zero. At $p = 1$ the reverse strict inequality holds.

Choose the balanced count vectors as N increases by incrementing one of their minimum coordinates. Each coefficient $W_n(j)$ is then nondecreasing. The coefficient of u^k is

$$W_n(k) = \frac{k!(k-1)}{2} \sum_i (n_i - 1) = \frac{k!(k-1)}{2} (N - k).$$

Indeed, exactly one run color appears twice; for each such color there are $(k+1)!/2 - k! = k!(k-1)/2$ proper arrangements of the multiset of run colors. This coefficient increases strictly with N , so the positive root decreases strictly with N . The substitution $p = 1/(1 + (d-1)u)$ proves the claimed monotonicity. If $N = k$, every coordinate equals one and $S_n(u) = k!u^{k-1}$, giving the initial value. Notice also that every root satisfies $u \leq (k!)^{-1/(k-1)} < 1$, so all crossings lie in the positive refresh regime used below. $\square$

Lemma 8.3 (positive integral and discrete comparison). *For fixed $k \geq 2$, define*

$$F_k(x) = \sum_{m_1, \dots, m_k \geq 1} M! \prod_{i=1}^k \frac{x^{m_i}}{m_i! (m_i - 1)!} = \int_0^\infty e^{-t} [\sqrt{xt} I_1(2\sqrt{xt})]^k dt, \quad M = \sum_i m_i. \quad (38)$$

Let n be a balanced positive count vector of total N , put $h = N/k$, and, for $0 < u < 1$, put $x = hu/(1-u)$. With $\mathcal{D} = x d/dx$,

$$0 \leq 1 - \frac{S_n(u)}{(1-u)^{N-1} h^{1-k} F_k(x)/x} \leq \frac{\mathcal{D}^2 F_k(x)}{h F_k(x)} = O_k\left(\frac{(1+x)^2}{h}\right). \quad (39)$$

In particular the relative error is $O_k((\log N)^2/N)$ when $x = O(\log N)$.

Proof. The equality in (38) follows from the Bessel power series and Tonelli's theorem, using $M! = \int_0^\infty e^{-t} t^M dt$. The integral is finite for every positive x : the bound $I_1(y) \leq e^y$ for $y \geq 0$ reduces its tail to a polynomial times $\exp(-t + 2k\sqrt{xt})$. Thus the positive series and all its termwise derivatives converge on compact subsets of $x > 0$.

Set $\sigma = p - r = p(1-u)$ and $v = r/\sigma = u/(1-u)$. Proposition 2.4, restricted to the positive coordinates, gives

$$S_n(u) = (1-u)^{N-1} \sum_{m_i \geq 1} M! v^{M-1} \prod_i \frac{\binom{n_i-1}{m_i-1}}{m_i!}. \quad (40)$$

Define

$$A_{i,m} = \frac{(m-1)!}{h^{m-1}} \binom{n_i-1}{m-1} = \prod_{j=1}^{m-1} \left(1 - \frac{h - n_i + j}{h}\right)_+.$$

Since $h - n_i > -1$, these are products of factors in $[0, 1]$, including beyond their finite support. The product-deficit inequality used in Theorem 5.2 gives

$$0 \leq 1 - \prod_i A_{i,m_i} \leq \frac{1}{h} \sum_i \frac{m_i(m_i+1)}{2} \leq \frac{M^2}{h}.$$

Replacing the binomial factors in (40) by $h^{m_i-1}/(m_i-1)!$ gives the denominator in (39). Summing the deficit bound proves its first inequality and the displayed explicit upper bound.

To bound the quotient of derivatives, for $x \geq 1$ use $M^2 \leq 3x^2(1+1/x)^M$. Indeed, $\log(1+1/x) \geq 1/(2x)$ and $\sup_{y \geq 0} y^2 e^{-y/2} = 16/e^2 < 3$. Positivity of the coefficients gives

$$\frac{\mathcal{D}^2 F_k(x)}{F_k(x)} \leq 3x^2 \frac{F_k(x+1)}{F_k(x)} = O_k(x^2),$$

where the last step follows from Lemma 8.4 below. For $0 < x \leq 1$, the quotient extends continuously to $x = 0$ with value k^2 , since $F_k(x) = k!x^k + O_k(x^{k+1})$. This proves the asserted bound without differentiating an asymptotic expansion. $\square$

Lemma 8.4 (Laplace expansion). *For each fixed $k \geq 2$, as $x \rightarrow \infty$,*

$$F_k(x) = \frac{k^{k/2+1}}{(4\pi)^{(k-1)/2}} x^{(k+1)/2} e^{k^2 x} \left(1 - \frac{k-1}{8kx} + O_k(x^{-2}) \right). \quad (41)$$

Proof. In (38) set $w = \sqrt{t}$ and $a = k\sqrt{x}$. The exponent becomes $-w^2 + 2k\sqrt{x}w = k^2x - (w-a)^2$. First restrict to $|w-a| \leq \sqrt{x}/2$. There $2\sqrt{x}w$ is bounded below by $(2k-1)x$, so the standard Bessel expansion [20, Eq. 10.40.1] applies uniformly:

$$[\sqrt{x}w I_1(2\sqrt{x}w)]^k = \frac{x^{k/4} w^{k/2} e^{2k\sqrt{x}w}}{(4\pi)^{k/2}} \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2}) \right).$$

On the complementary set, the elementary exponential bound for I_1 dominates the integrand by $2x^{k/2} w^{k+1} e^{k^2x - (w-a)^2}$. Gaussian tail estimates therefore show that its integral is at most $e^{k^2x - cx}$ times a fixed power of x , for some $c > 0$. It is exponentially smaller than the main term.

Put $\alpha = k/2 + 1$. Apart from that negligible tail, the integral is

$$\frac{2x^{k/4} e^{k^2x}}{(4\pi)^{k/2}} \int_{|w-a| \leq \sqrt{x}/2} e^{-(w-a)^2} w^\alpha \left(1 - \frac{3k}{16\sqrt{x}w} + O_k(x^{-2}) \right) dw.$$

Taylor-expand w^α about a . The odd terms integrate to zero on this symmetric interval, the quadratic term contributes $\alpha(\alpha-1)/(4a^2)$ relatively, and the fourth-order remainder is $O_k(a^{-4})$: the interval stays a fixed positive fraction of a away from zero and all Gaussian moments are finite. Extending the centered Gaussian integrals to the real line changes them exponentially little. For the term involving $w^{\alpha-1}/\sqrt{x}$, keeping its constant term suffices, because its quadratic correction has relative order x^{-2} . Since $\int e^{-y^2} dy = \sqrt{\pi}$ and $\int y^2 e^{-y^2} dy = \sqrt{\pi}/2$, the total relative correction is

$$\frac{\alpha(\alpha-1)}{4k^2x} - \frac{3}{16x} = -\frac{k-1}{8kx}.$$

The leading constant simplifies to the one in (41), completing the proof. $\square$

Proof of the asymptotic assertions in Theorems 8.1 and 8.2. Set $s = k-1$, $z = Nu$, and $D_k = k^{k+1/2}/(4\pi)^{(k-1)/2}$. The preceding lemmas imply, uniformly on $c \log N \leq z \leq C \log N$ for fixed $0 < c < C$,

$$S_n(z/N) = D_k \frac{z^{s/2} e^{sz}}{N^s} \left(1 - \frac{s}{8z} + O_{k,c,C}(z^{-2} + z^2/N) \right). \quad (42)$$

Indeed, $x = z/(k(1-u))$, and

$$k^2x + (N-1) \log(1-u) - sz = O_k(z^2/N).$$

The prefactor changes by $1 + O_k(z/N)$ on replacing x by z/k ; rounding of the balanced coordinates is covered by (39).

At $z = (1 - \varepsilon) \log N$ the ratio in (42) tends to zero, whereas at $z = (1 + \varepsilon) \log N$ it tends to infinity. The unique root therefore has $z/\log N \rightarrow 1$. Taking logarithms at the root and dividing by s gives

$$z + \frac{1}{2} \log z = L + a_k + \frac{1}{8z} + O_k(z^{-2} + z^2/N).$$

Successive substitution, or inversion as in Theorem 5.2, gives (36). The conversion to physical persistence uses $N(1 - p) = (d - 1)z/(1 + (d - 1)z/N) = (d - 1)z + O_d(z^2/N)$. Substituting $z = L - \frac{1}{2} \log L + t$ in (42) proves (37), uniformly on compact t intervals.

For the ordering of constants, extend $g(x) = (x + 1/2) \log x / (x - 1)$ to $x \geq 2$. The sign of $g'(x)$ is the sign of $H(x) = 2x^2 - x - 1 - 3x \log x$. Here $H(2) = 5 - 6 \log 2 > 0$, and $H'(x) = 4x - 4 - 3 \log x > 0$ for $x \geq 2$, since its derivative is $4 - 3/x > 0$ and $H'(2) > 0$. Thus g increases strictly, so a_k decreases strictly. $\square$

8.1 Thresholds throughout the interior of a face

The same argument gives a spatially varying threshold throughout a face, not just at its center. All constants in the following statement use the *actual* fractions $\alpha_i = n_i/N$. This is necessary if the count vector varies with N without a specified rate of convergence.

Theorem 8.5 (interior composition thresholds). *Fix $k \geq 2$ and $\varepsilon > 0$. Let $n_1 + \dots + n_k = N$, with $\alpha_i = n_i/N \geq \varepsilon$, and define*

$$A = \sum_{i=1}^k \sqrt{\alpha_i}, \quad g = A^2 - 1, \quad s = k - 1, \quad D(\alpha) = \frac{A^{k/2+1} (\prod_i \alpha_i)^{-3/4}}{(4\pi)^{s/2}}.$$

Uniformly for these count vectors and $c \log N \leq z \leq C \log N$, with fixed $0 < c < C$,

$$S_n(z/N) = D(\alpha) \frac{z^{s/2} e^{gz}}{N^s} (1 + O_{k,\varepsilon,c,C}(z^{-1} + z^2/N)). \quad (43)$$

Let $u_n > 0$ be the unique root of $S_n(u) = 1$. Writing $L = \log N$,

$$Nu_n = \frac{s}{g} L - \frac{s}{2g} \log L - \frac{\log D(\alpha) + (s/2) \log(s/g)}{g} + O_{k,\varepsilon} \left(\frac{\log L}{L} + \frac{L^2}{N} \right). \quad (44)$$

These estimates are uniform on every compact subset of the relative interior of the face. For a fixed target α and integer counts $n_i = N\alpha_i + O(1)$, the constants may equivalently be evaluated at that target, without changing the error order.

Proof. The proper-run polynomial used above has nonnegative coefficients and first coefficient $k!u^{k-1}$ for every positive count vector, which gives the unique positive root. To obtain the approximation, put $v = u/(1 - u)$ and $y = Nv$, and introduce

$$G_\alpha(y) = \int_0^\infty e^{-t} \prod_i [\sqrt{\alpha_i y t} I_1(2\sqrt{\alpha_i y t})] dt.$$

Replacing each binomial coefficient in (40) by $n_i^{m_i-1}/(m_i - 1)!$ gives

$$S_n(u) = \frac{(1 - u)^{N-1}}{v \prod_i n_i} G_\alpha(y) (1 + O_{k,\varepsilon}((1 + y)^2/N)). \quad (*)$$

Here is a uniform justification of the error. The corresponding falling-factorial deficit is bounded by $\sum_i m_i(m_i - 1)/(2n_i) \leq M^2/(2\varepsilon N)$. The positive-series argument in Lemma 8.3 bounds the weighted second moment of M by a constant times $(1 + y)^2$. Uniformity follows from the uniform saddle expansion below, which bounds $G_\alpha(y + 1)/G_\alpha(y)$ on the compact set $\{\alpha_i \geq \varepsilon, \sum_i \alpha_i = 1\}$; for bounded y , continuity and the positive leading coefficient give the same conclusion.

Set $w = \sqrt{t}$. The exponential factor of the integrand is

$$\exp(-w^2 + 2A\sqrt{y}w) = \exp(A^2y - (w - A\sqrt{y})^2).$$

Every α_i stays bounded away from zero, so the Bessel expansions, Gaussian tail estimates, and Taylor remainders in Lemma 8.4 are uniform here. Keeping their leading terms yields

$$G_\alpha(y) = \frac{A^{k/2+1}(\prod_i \alpha_i)^{1/4}}{(4\pi)^{s/2}} y^{(k+1)/2} e^{A^2y} (1 + O_{k,\varepsilon}(y^{-1})).$$

Substitute this into (*) and use $v \prod_i n_i = yN^{k-1} \prod_i \alpha_i$. For $u = z/N$ in the stated range, $y = z + O(z^2/N)$ and $(N - 1)\log(1 - u) = -z + O(z^2/N)$. This proves (43) with the specified constant.

On the compact set of compositions, both g and $D(\alpha)$ have positive lower and finite upper bounds. Testing the ratio at suitable constant multiples of $\log N$ brackets the unique root uniformly. Taking logarithms there gives

$$gz + \frac{s}{2} \log z = sL - \log D(\alpha) + O_{k,\varepsilon}(z^{-1} + z^2/N).$$

First $z = (s/g)L + O_{k,\varepsilon}(\log L)$; one substitution then gives (44). The derivative of its left-hand side is at least g , bounded uniformly away from zero, so the stated error is preserved on inversion. Finally, the functions of α appearing in the formula have bounded derivatives on a slightly smaller compact set; changing α by $O(1/N)$ changes the expression by $O(L/N)$, which is absorbed by its error term. $\square$

At a balanced composition, $A = \sqrt{k}$, $g = k - 1$, and $D(\alpha) = D_k$, recovering the preceding center formula through its constant term. For $k = 2$, $g = 2\sqrt{\alpha_1\alpha_2}$, recovering the binary bulk scale. Cauchy–Schwarz gives $g \leq k - 1$, with equality only at the balanced composition, so the leading threshold coefficient s/g is minimized exactly at the center of each face.

8.2 Balanced bins and global modes

Theorem 8.6 (balancing within a support). *For $1/d \leq p < 1$, among bins with a prescribed set of positive coordinates and total N , probability is maximized precisely by the balanced count vectors. More precisely, transferring one unit from a coordinate b to another coordinate a , where $b \geq a + 2$ and $a \geq 1$, strictly increases the bin probability.*

Proof. At $p = 1/d$ this follows from the ordinary multinomial formula. For $1/d < p < 1$, put $\sigma = p - r > 0$ and $v = r/\sigma > 0$. The positive refresh sum has the integral form

$$P_N(n) = \frac{\sigma^{N-1}}{dv} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt, \quad B_n(y) = \sum_{m=1}^n \binom{n-1}{m-1} \frac{y^m}{m!}. \quad (45)$$

For each $y > 0$, the sequence $B_n(y)$, $n \geq 1$, is strictly log-concave. We give an elementary proof of this instance of preservation of log-concavity by the binomial transform. Write

$$\frac{B_n(y)}{y} = b_{n-1}, \quad b_j = \sum_{i=0}^j \binom{j}{i} a_i, \quad a_i = \frac{y^i}{(i+1)!}, \quad c_j = \sum_{i=0}^j \binom{j}{i} a_{i+1}.$$

Pascal's identity gives $b_{j+1} = b_j + c_j$. The ratio c_j/b_j is the expectation of the strictly decreasing sequence $a_{i+1}/a_i = y/(i+2)$ under weights proportional to $\binom{j}{i}a_i$. When j increases to $j+1$, the likelihood ratio of the new and old unnormalized weights on the old support is $(j+1)/(j+1-i)$, increasing in i ; additionally a strictly positive weight appears at $i = j+1$. Thus these weights shift strictly toward larger indices. The expectation of a strictly decreasing sequence decreases strictly. This last implication follows directly by cross-multiplying the expectations and pairing terms with indices $i < \ell$, so no limiting probability argument is involved. Hence $b_{j+1}/b_j = 1 + c_j/b_j$ decreases strictly with j , proving the asserted strict log-concavity.

It follows that $B_{a+1}(y)B_{b-1}(y) > B_a(y)B_b(y)$ whenever $b \geq a+2$. Apply this inequality pointwise for $y = vt > 0$ in (45) and integrate. Repeated transfers give the conclusion. $\square$

Corollary 8.7 (global modes in the critical window). *Fix d . In the window of Theorem 8.2, if $t < a_d$, then for all sufficiently large N the only global maxima are the d vertices. If $t > a_d$, then the only global maxima are the fully supported balanced bins. These assertions are uniform on compact sets of t separated from a_d . At the exact full-center/vertex crossing, for all sufficiently large N the global maxima are exactly the vertices and the fully supported balanced bins, with both classes present.*

Proof. Theorem 8.6 reduces the possible modes to balanced face bins of support sizes $1, \dots, d$. For $2 \leq k \leq d$ their limiting log ratios to a vertex are $(k-1)(t-a_k)$. If $t < a_d$, these are all negative. If $t > a_d$, the full-center ratio exceeds the vertex ratio, and for $2 \leq k < d$ its limiting log ratio exceeds the k -face one by

$$(d-k)(t-a_d) + (k-1)(a_k-a_d) > 0.$$

Uniformity in Theorem 8.2 gives both assertions. At the exact full-center crossing, its window parameter tends to a_d by Theorem 8.1; every proper face-center ratio then has a limit strictly less than one. The same reduction proves the final claim. $\square$

Example 8.8 (an intermediate face can be a finite-size mode). The preceding asymptotic restriction cannot be replaced by a claim for every N . Take $d = 3$, $N = 4$, and $p = 10/17$, so $u = 7/20$. For an edge center and a full center respectively,

$$\frac{P_4(2, 2, 0)}{V_4} = 2u + 2u^2 + 2u^3 = \frac{4123}{4000} > 1, \quad \frac{P_4(2, 1, 1)}{V_4} = 6u^2 + 6u^3 = \frac{3969}{4000} < 1.$$

Theorem 8.6 shows that the three edge centers are the only global modes. Thus the common asymptotic window coexists with additional finite-size modal behavior.

9 A uniform simplex estimate and a boundary crossover

The compact-interior assumption in Theorem 8.5 excludes bins approaching a lower-dimensional face. We now remove that assumption from the discrete-to-Bessel estimate and resolve a transition

in which several positive coordinates become small. The asymptotic transition proved below has at least two macroscopic coordinates; the one-macroscopic-coordinate case has a different persistence scale.

Put

$$\Phi(x) = \sum_{j \geq 0} \frac{x^j}{j!(j+1)!} = \frac{I_1(2\sqrt{x})}{\sqrt{x}}, \quad \Phi(0) = 1.$$

As before, $S_n(u)$ is the probability of the positive count vector n divided by the probability of one vertex, and $u = r/p$. All the statements are independent of the ambient number of colors once the support is fixed.

Lemma 9.1 (a uniform Laguerre–Bessel inequality). *For every integer $a \geq 1$ and real $w > 0$,*

$$0 \leq 1 - \frac{B_a(w)}{w\Phi(aw)} \leq \frac{w}{2}, \quad B_a(w) = \sum_{m=1}^a \binom{a-1}{m-1} \frac{w^m}{m!}. \quad (46)$$

Proof. The coefficient of $(aw)^j/[j!(j+1)!]$ in $B_a(w)/w$ is $\prod_{h=1}^j (1 - h/a)_+$. Its deficit from one lies between zero and $j(j+1)/(2a)$, including for $j \geq a$. Consequently

$$0 \leq \Phi(aw) - \frac{B_a(w)}{w} \leq \frac{1}{2a} \sum_{j \geq 0} \frac{j(j+1)(aw)^j}{j!(j+1)!} = \frac{w}{2} \Phi(aw).$$

□

Theorem 9.2 (a uniform Bessel majorant on every simplex stratum). *Let $n_1, \dots, n_k$ be arbitrary positive integers with sum N , and let $0 < u < 1$, $v = u/(1-u)$. Define*

$$\mathcal{Q}_n(u) = (1-u)^{N-1} v^{k-1} \int_0^\infty e^{-t} t^k \prod_{i=1}^k \Phi(n_i vt) dt.$$

Then, with no lower bound on any n_i/N ,

$$0 \leq 1 - \frac{S_n(u)}{\mathcal{Q}_n(u)} \leq k(k+1)v + \frac{k}{2}v^2 \left(\sum_i \sqrt{n_i} \right)^2 \leq k(k+1)v + \frac{k^2}{2}Nv^2. \quad (47)$$

Thus this approximation has relative error $O_k((\log N)^2/N)$ whenever $u = O(\log N/N)$, uniformly over every positive composition of N .

Proof. The positive refresh formula gives

$$S_n(u) = \frac{(1-u)^{N-1}}{v} \int_0^\infty e^{-t} \prod_i B_{n_i}(vt) dt.$$

Applying Lemma 9.1 to each factor, and then the product-deficit inequality, bounds the relative deficit by $(kv/2)\mathbb{E}[T]$, where T has density proportional to $e^{-t}t^k \prod_i \Phi(n_i vt)$. For the probability weights proportional to $x^j/[j!(j+1)!]$, the identity $\mathbb{E}[J(J+1)] = x$ implies

$$x \frac{\Phi'(x)}{\Phi(x)} = \mathbb{E}[J] \leq \sqrt{x}.$$

Integration by parts therefore gives

$$\mathbb{E}[T] = k + 1 + \sum_i \mathbb{E} \left[\frac{n_i v T \Phi'(n_i v T)}{\Phi(n_i v T)} \right] \leq k + 1 + \sqrt{v} \left(\sum_i \sqrt{n_i} \right) \sqrt{\mathbb{E}[T]}.$$

The boundary terms vanish: near zero the density is $O(t^k)$, while $\Phi(x) \leq e^{2\sqrt{x}}$ bounds the tail by a polynomial times $\exp(-t + 2\sqrt{vt} \sum_i \sqrt{n_i})$. This also proves finiteness of the integral and of every moment used above. Applying $2ab \leq a^2 + b^2$ to the final term yields $\mathbb{E}[T] \leq 2(k+1) + v(\sum_i \sqrt{n_i})^2$. The claimed estimates follow, the last one by Cauchy–Schwarz. $\square$

9.1 Coordinates of order $N/(\log N)^2$

Fix integers $r \geq 2$ and $\ell \geq 1$, and put $k = r + \ell$. Consider r large counts $n_1, \dots, n_r$ and ℓ further positive counts $a_1, \dots, a_\ell$. Write

$$N = \sum_i n_i + \sum_j a_j, \quad M = \sum_i n_i, \quad L = \log N, \quad \beta_i = \frac{n_i}{M}, \quad \rho_j = \frac{a_j L^2}{N}.$$

The β_i and ρ_j below always refer to these actual counts. Define

$$A = \sum_{i=1}^r \sqrt{\beta_i}, \quad g = A^2 - 1, \quad K = r + \ell - 1, \quad H = \frac{r-1}{2} + 2\ell, \quad c = \frac{K}{g},$$

and

$$D_r(\beta) = \frac{A^{r/2+1} (\prod_i \beta_i)^{-3/4}}{(4\pi)^{(r-1)/2}}.$$

Theorem 9.3 (a crossover between neighboring simplex strata). *Fix $\varepsilon > 0$ and $R < \infty$. Assume $\beta_i \geq \varepsilon$ and $0 < \rho_j \leq R$. Uniformly for such count vectors and $c_0 L \leq z \leq C_0 L$, where $0 < c_0 < C_0 < \infty$ are fixed,*

$$S_{(n,a)}(z/N) = D_r(\beta) A^{2\ell} \frac{z^H e^{gz}}{N^K} \prod_{j=1}^{\ell} \Phi\left(\frac{A^2 a_j z^2}{N}\right) \left(1 + O\left(L^{-1} + \frac{L^2}{N}\right)\right). \quad (48)$$

The unique positive solution $u_{n,a}$ of $S_{(n,a)}(u) = 1$ consequently satisfies

$$\begin{aligned} Nu_{n,a} = cL - \frac{H}{g} \log L - \frac{1}{g} \left\{ \log(D_r(\beta) A^{2\ell}) + H \log c + \sum_{j=1}^{\ell} \log \Phi(A^2 c^2 \rho_j) \right\} \\ + O\left(\frac{\log L}{L} + \frac{L^2}{N}\right). \end{aligned} \quad (49)$$

The constants in the errors depend only on the displayed fixed parameters.

Proof. Let $v = u/(1-u)$ and $Y = Mv$. By Theorem 9.2, with relative error $O_k(v + Nv^2)$, the exact ratio equals

$$\frac{(1-u)^{N-1} v^{\ell-1}}{\prod_{i=1}^r n_i} \int_0^\infty e^{-t} t^\ell \prod_{i=1}^r \left[\sqrt{\beta_i Y t} I_1(2\sqrt{\beta_i Y t}) \right] \prod_{j=1}^{\ell} \Phi(a_j v t) dt. \quad (50)$$

In the stated regime $Y \asymp L$ and all $a_j v Y$ are bounded. Set $w = \sqrt{t}$. The exponential from the r large-coordinate factors is $\exp(A^2 Y - (w - A\sqrt{Y})^2)$. On $|w - A\sqrt{Y}| \leq \sqrt{Y}/2$, the large-coordinate Bessel arguments are bounded below by a positive constant times Y . Their usual large-argument expansions are uniform, with relative error $O(1/Y)$. The remaining smooth factor, apart from constants and powers of Y , is

$$w^{r/2+1+2\ell} \prod_j \Phi(a_j v w^2).$$

Divide this factor by its value at $w = A\sqrt{Y}$. On the same interval its second derivative is $O(1/Y)$, uniformly: $w/\sqrt{Y}$ lies in a fixed compact subset of $(0, \infty)$, the quantities $a_j v Y$ lie in a fixed compact interval, and $\Phi \geq 1$ there. Taylor's theorem, followed by integration against the centered Gaussian, eliminates the linear term and gives relative error $O(1/Y)$.

For completeness, the part outside this interval is exponentially negligible uniformly. The bounds $I_1(x) \leq e^x$ and $\Phi(x) \leq e^{2\sqrt{x}}$ give a polynomial times

$$\exp\left(A^2 Y - (w - A\sqrt{Y})^2 + 2w \sum_j \sqrt{a_j v}\right).$$

Here $\sum_j \sqrt{a_j v} = O(Y^{-1/2})$, so completing the square leaves a Gaussian centered only $O(Y^{-1/2})$ away from $A\sqrt{Y}$. On the excluded region its integral is exponentially small relative to the main term, even after the polynomial factors are included. The same argument also justifies extending the central Gaussian integrals to the full real line.

Evaluation of the leading Gaussian integral in (50) now gives

$$\begin{aligned} S_{(n,a)}(u) &= D_r(\beta) A^{2\ell} \frac{Y^H}{M^K} e^{A^2 Y + (N-1) \log(1-u)} \prod_j \Phi(A^2 a_j M v^2) \\ &\quad \times (1 + O(Y^{-1} + v + N v^2)). \end{aligned}$$

Since $\sum_j a_j/N = O(L^{-2})$, at $u = z/N$ we have

$$Y = z + O(L^{-1} + L^2/N), \quad A^2 Y + (N-1) \log(1-u) = g z + O(L^{-1} + L^2/N).$$

Replacing Y by z , M by N , and $A^2 a_j M v^2$ by $A^2 a_j z^2/N$ in the other factors changes the expression by a relative $O(L^{-2} + L/N)$, because the arguments of Φ stay bounded and $\Phi \geq 1$. This proves (48).

The proper-run polynomial has nonnegative coefficients, zero constant coefficient, and positive coefficient $k!$ at u^{k-1} ; hence the crossing is unique. On the compact set of permitted β , both g and $D_r(\beta)$ are bounded above and bounded away from zero. Testing (48) at fixed sufficiently small and large multiples of L brackets the root uniformly. Its logarithm gives

$$g z + H \log z = K L - \log(D_r(\beta) A^{2\ell}) - \sum_j \log \Phi(A^2 \rho_j (z/L)^2) + O(L^{-1} + L^2/N).$$

First $z = cL + O(\log L)$, so the sum of logarithms on the right differs from its value at $z = cL$ by $O(\log L/L)$. Substituting this estimate into $\log z$ proves (49); inversion preserves the error because the derivative of $g z + H \log z$ is at least $g > 0$. $\square$

Corollary 9.4 (invisibility of subcritical rare counts). *Under the hypotheses of Theorem 9.3, if all $a_j = o(N/(\log N)^2)$, the sum involving Φ in (49) is $o(1)$. Thus, at fixed actual large composition β , the rare counts do not affect the expansion through its constant term. Their number ℓ still changes both its leading and its logarithmic coefficients.*

The scale $N/(\log N)^2$ here concerns approaching a simplex face with at least two macroscopic coordinates. It is not a replacement for the binary edge scale of Section 6: there is only one macroscopic coordinate at a binary endpoint, and then $g = 0$. Nor does Theorem 9.3 assert an error bound when any ρ_j is unbounded.

Lemma 9.5 (exact insertion of a singleton color). *Let n have total size M and let $\mathcal{D} = u d/du$. Then*

$$S_{(n,1)}(u) = [2u + (M-1)u^2]S_n(u) + (u-u^2)\mathcal{D}S_n(u).$$

Proof. Insert the new color into a word of length M with j changes. The two end positions contribute a factor $2u$; the j change gaps contribute ju , and the $M-1-j$ repeat gaps contribute $(M-1-j)u^2$. Removing the unique new letter reverses this construction. Summing the resulting identity over all words proves the formula. $\square$

The singleton identity gives an exact finite-size check on the factor $A^2 z^2/N$ contributed by one very small coordinate in (48). The finite-size inequalities above are proved algebraically; the asymptotic statements use real analysis and are not included in the accompanying Lean formalization.

9.2 Fixed rare counts near a vertex

The singular case of one large coordinate admits a different, finite polynomial limit. In this result the rare counts are fixed, and the large coordinate b , rather than the total count, is the asymptotic parameter.

Theorem 9.6 (a multicolor Laguerre limit at a vertex). *Fix $\ell \geq 1$ and positive integers $a_1, \dots, a_\ell$. Define*

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad Q(y) = \sum_{i < j} B'_{a_i}(y) B'_{a_j}(y) \prod_{h \neq i,j} B_{a_h}(y).$$

The sum defining Q is zero when $\ell = 1$. Uniformly on compact subsets of $y > 0$, as $b \rightarrow \infty$,

$$S_{(b,a)}(\sqrt{y/b}) = H(y) + \frac{2\sqrt{y}}{\sqrt{b}}(H'(y) + yQ(y)) + O_a(b^{-1}). \quad (51)$$

Let $y_a > 0$ be the unique root of $H(y_a) = 1$, and put $c_a = \sqrt{y_a}$. The unique endpoint-crossing odds satisfy

$$u_{b,a} = \frac{c_a}{\sqrt{b}} - \frac{1 + y_a Q(y_a)/H'(y_a)}{b} + O_a(b^{-3/2}). \quad (52)$$

For one rare color this recovers the binary fixed-edge correction $-1/b$. For at least two rare colors the coefficient of $1/b$ is strictly less than -1 .

Proof. Fix rare-run counts $m_i \in \{1, \dots, a_i\}$, put $m = \sum_i m_i$, and let q be the number of runs of the large color. The rare run lengths can be chosen in $\prod_i \binom{a_i-1}{m_i-1}$ ways. Since successive large-color runs must be separated, $q \leq m+1$; the large run lengths have $\binom{b-1}{q-1}$ choices. At $u = \sqrt{y/b}$, configurations with these run counts contribute $O_a(b^{(q-m-1)/2})$. There are finitely many possible m_i, q , independently of b , so only $q = m+1$ and $q = m$ affect the expansion through order $b^{-1/2}$.

If $q = m+1$, the run word starts and ends in the large color and alternates between large and rare colors. There are $m!/\prod_i m_i!$ rare orders. Using $\binom{b-1}{m} = b^m/m! + O_a(b^{m-1})$ and summing gives $H(y)$, with an $O_a(b^{-1})$ error.

If $q = m$, there are two possibilities. One endpoint is rare and the run word otherwise alternates, giving $2m!/\prod_i m_i!$ orders. These terms sum to $2\sqrt{y}H'(y)/\sqrt{b}$. Otherwise both endpoints are large and exactly one pair of consecutive rare runs is present. The members of that pair must have different colors. For an ordered pair of colors i, j , contracting the marked adjacent pair shows that there are $(m-1)!/[(m_i-1)!(m_j-1)!\prod_{h \neq i,j} m_h!]$ orders. Summing over the two orientations and all $i < j$ gives $2y^{3/2}Q(y)/\sqrt{b}$. Terms with $q \leq m-1$ and the next terms of the binomial expansions are $O_a(b^{-1})$, uniformly for y in a positive compact set. This proves (51).

The polynomial H has positive coefficients, tends to zero at zero, and grows to infinity, so y_a exists uniquely and $H'(y_a) > 0$. The same positivity argument gives the unique crossing for each b . First bracket it using (51) to obtain $bu_{b,a}^2 \rightarrow y_a$. The finite run expansion may also be differentiated uniformly on this compact neighborhood; its derivative tends to $H'(y_a) > 0$. Substitution of $u = c_a/\sqrt{b} + \alpha/b$ into (51) then forces $\alpha = -1 - y_a Q(y_a)/H'(y_a)$, and the uniform remainder gives (52). If $\ell \geq 2$, all terms in $Q(y)$ are positive for $y > 0$, proving the strict inequality. $\square$

If all ℓ rare colors occur once, then $H(y) = y^\ell$ and $y_a = 1$, giving the explicit specialization

$$u_{b,1,\dots,1} = b^{-1/2} - \frac{\ell+1}{2b} + O_\ell(b^{-3/2}).$$

An independent exact formula in this case is

$$S_{(b,1,\dots,1)}(u) = \ell! \sum_{q=1}^{\min(b,\ell+1)} \binom{\ell+1}{q} \binom{b-1}{q-1} u^{q+\ell-1}.$$

Indeed, order the ℓ distinct rare letters, choose the q gaps among their $\ell+1$ end and interior gaps that contain large-color runs, and choose the positive lengths of those runs. In particular two singleton rare colors give

$$u_{b,1,1} = b^{-1/2} - \frac{3}{2}b^{-1} + O(b^{-3/2}).$$

This agrees with the exact polynomial $S_{(b,1,1)}(u) = 6u^2 + 6(b-1)u^3 + (b-1)(b-2)u^4$. Theorem 9.6 does not assert uniformity when the rare counts grow with b . Section 10 resolves the leading crossing uniformly for total rare count $A = o(b)$; uniform higher-order corrections in that full regime remain open here.

10 Uniform completion of the near-vertex regime

The fixed-count hypothesis in Theorem 9.6 can be removed at the level of the leading crossing threshold. The result is uniform over every rare composition with total mass $o(b)$. An accompanying example explains why the corresponding probability approximation need not be uniform: its error can diverge even while the relative error of the crossing threshold tends to zero.

Throughout this section, $\ell \geq 1$ is fixed, $a_1, \dots, a_\ell$ are positive integers, $A = \sum_i a_i$, and

$$H(y) = \prod_{i=1}^{\ell} B_{a_i}(y), \quad \kappa(y) = \frac{yH'(y)}{H(y)}.$$

Write y_a for the unique positive solution of $H(y_a) = 1$.

Lemma 10.1 (elasticity and concavity of the rare-count polynomial). *The polynomial H is convex, $\log H$ is concave on $(0, \infty)$, and κ is nondecreasing. Moreover,*

$$\frac{1}{A+1} \leq y_a \leq 1, \quad \kappa(y) \leq \ell + \sqrt{\ell A y}, \quad \kappa(y) \geq \max\{\ell, \tfrac{1}{2}\sqrt{A y}\} \quad (0 < y \leq 1). \quad (53)$$

Proof. Convexity follows directly from the nonnegative coefficients. For one factor, give $J \in \{0, \dots, a-1\}$ probabilities proportional to

$$r_j = \binom{a-1}{j} \frac{y^j}{(j+1)!}.$$

The successive ratio $R(j) = (j+1)r_{j+1}/r_j = y(a-1-j)/(j+2)$, with $R(a-1) = 0$, is nonincreasing. Summing the ratio identity gives $\mathbb{E}J = \mathbb{E}R(J)$ and $\mathbb{E}J^2 = \mathbb{E}[(J+1)R(J)]$. Therefore

$$\text{Var } J = \mathbb{E}J + \text{Cov}(J, R(J)) \leq \mathbb{E}J.$$

The covariance is nonpositive because, for an independent copy J' , twice the covariance equals $\mathbb{E}[(J - J')(R(J) - R(J'))] \leq 0$. Putting $M = J + 1$ and $\kappa_a = yB'_a/B_a = \mathbb{E}M$, we obtain

$$(\log B_a)'' = \frac{\text{Var } M - \mathbb{E}M}{y^2} \leq 0, \quad y\kappa'_a = \text{Var } M \geq 0.$$

This proves both qualitative assertions without invoking the locations of the zeros of a Laguerre polynomial.

The elementary differential equation $yB''_a + yB'_a = aB_a$ gives

$$ay = \kappa_a^2 + \text{Var } M + (y-1)\kappa_a.$$

Using $0 \leq \text{Var } M \leq \kappa_a - 1$ yields $\kappa_a \leq 1 + \sqrt{ay}$ and $\kappa_a^2 + y\kappa_a \geq ay + 1$. For $0 < y \leq 1$, these imply $\kappa_a \geq \max\{1, \sqrt{ay} - 1/2\} \geq \frac{1}{2}\sqrt{ay}$. Summing over the factors and applying Cauchy–Schwarz proves the elasticity bounds. Finally, $B_a(1) \geq 1$ and

$$B_a(y) \leq y\Phi(ay) \leq ye^{ay/2},$$

where $(j+1)! \geq 2^j$ proves the last inequality term by term. Thus $H(1) \geq 1$, whereas $H(1/(A+1)) \leq (A+1)^{-\ell} e^{A/(2(A+1))} < 1$. $\square$

Lemma 10.2 (an exact compound-binomial representation). *For $0 < u < 1$, put $v = u/(1-u)$. Let G have the gamma distribution with shape 2 and rate 1, let $E_1, \dots, E_{b-1}$ be independent rate-one exponential variables, and let $Z_1, \dots, Z_{b-1}$ be independent Bernoulli variables with parameter u . Take all these variables independent and set*

$$T = G + \sum_{j=1}^{b-1} Z_j E_j.$$

Then

$$S_{(b,a)}(u) = (1-u)^A \mathbb{E}[H(vT)], \quad \mathbb{E}e^{sT} = (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1} \quad (s < 1). \quad (54)$$

In particular, set $y = bu^2$, $\kappa = \kappa(y)$ and $s = \kappa/[bu(1-u)]$. Whenever $s < 1$,

$$(1-u)^A H(y) \leq S_{(b,a)}(u), \quad (55)$$

$$S_{(b,a)}(u) \leq (1-u)^A H(y) e^{-\kappa} (1-s)^{-2} \left(1 + \frac{us}{1-s}\right)^{b-1}. \quad (56)$$

Proof. The exact positive-refresh integral is

$$S_{(b,a)}(u) = \frac{(1-u)^{b+A-1}}{v} \int_0^\infty e^{-t} B_b(vt) H(vt) dt.$$

The integral of $e^{-t} B_b(vt)$ is $v(1+v)^{b-1}$. After normalization its density is a mixture of gamma densities with shape $J+2$, where $J \sim \text{Bin}(b-1, u)$. This is exactly the distribution of T , proving the identity and its moment generating function. Since $\mathbb{E}T = 2 + (b-1)u$, $v\mathbb{E}T \geq bu^2 = y$. Convexity and monotonicity of H prove (55). Concavity of $\log H$ gives, for $w > 0$,

$$H(w) \leq H(y) \exp\left(\frac{\kappa}{y}(w-y)\right).$$

Insert $w = vT$ and use the moment generating function to prove (56). $\square$

Theorem 10.3 (all growing rare counts near a vertex). *For fixed ℓ , uniformly over all positive rare counts with $A = o(b)$, the unique positive crossing odds satisfy*

$$u_{b,a} = \sqrt{\frac{y_a}{b}} \left(1 + O_\ell\left(\sqrt{\frac{A+1}{b}}\right)\right), \quad \prod_i B_{a_i}(y_a) = 1. \quad (57)$$

A quantitative, unoptimized version is as follows. Put

$$K = \ell + \sqrt{\ell} + 1, \quad \varepsilon = \sqrt{\frac{A+1}{b}}, \quad \delta = 16K\varepsilon.$$

If $\delta \leq 1/2$, then

$$(1-\delta)\sqrt{y_a/b} \leq u_{b,a} \leq (1+\delta)\sqrt{y_a/b}. \quad (58)$$

No restriction such as $A(\log A)^2 = o(b)$ is needed.

Proof. The proper-run polynomial has positive coefficients, zero constant coefficient, and is unbounded at infinity; its crossing is unique. Put $u_0 = \sqrt{y_a/b}$ and $\kappa_0 = \kappa(y_a)$. At $u_+ = (1+\delta)u_0$ we have $u_+ < 1/4$. Monotonicity of κ and the lower bound in Lemma 10.1 give

$$\log H(bu_+^2) \geq 2\kappa_0 \log(1+\delta), \quad \frac{Au_0}{\kappa_0} \leq 2\sqrt{A/b} \leq 2\varepsilon.$$

Consequently, by (55),

$$\log S(u_+) \geq 2\kappa_0 \log(1+\delta) - \frac{Au_+}{1-u_+} \geq \kappa_0 \left(\frac{4}{3}\delta - 4\varepsilon\right) > 0.$$

At $u_- = (1-\delta)u_0$, write $y = bu_-^2$ and $\kappa = \kappa(y)$. The inequalities $y_a \geq 1/(A+1)$, $\delta \leq 1/2$, and (53) imply

$$u_- \leq \varepsilon, \quad bu_- \geq \frac{1}{2\varepsilon}, \quad s = \frac{\kappa}{bu_-(1-u_-)} \leq 2(2\ell + \sqrt{\ell})\varepsilon \leq 4K\varepsilon \leq \frac{1}{8}.$$

Using $\log(1+x) \leq x$, $-\log(1-s) \leq s/(1-s)$ and $u_-, s \leq 1/4$ in (56), and dropping the nonpositive term $A \log(1-u_-)$, gives

$$\begin{aligned} \log \frac{S(u_-)}{H(y)} &\leq -\kappa + \frac{2s}{1-s} + \frac{bu_-s}{1-s} \\ &= \kappa \left(\frac{1}{(1-u_-)(1-s)} - 1 \right) + \frac{2s}{1-s} \\ &\leq 5\kappa(u_- + s) \leq 25K\kappa\varepsilon. \end{aligned}$$

Here $\kappa \geq \ell \geq 1$ was used in the last two inequalities. Again by monotonicity of κ ,

$$\log H(y) \leq 2\kappa \log(1 - \delta) \leq -2\kappa\delta.$$

Thus $\log S(u_-) \leq \kappa(-32K + 25K)\varepsilon < 0$. The two strict inequalities and uniqueness bracket the crossing. $\square$

Corollary 10.4 (an explicit shape law for growing rare counts). *Suppose $A \rightarrow \infty$, $A = o(b)$, and the actual proportions $\alpha_i = a_i/A$ stay in a fixed compact subset of the positive ℓ -simplex. Put $h = \sum_i \sqrt{\alpha_i}$. Then*

$$u_{b,a} \sim \frac{\ell \log A}{2h\sqrt{Ab}}.$$

More precisely, the polynomial root satisfies

$$2h\sqrt{Ay_a} = \ell \log A - \frac{\ell}{2} \log \log A + \frac{\ell}{2} \log \left(\frac{8\pi h}{\ell} \right) + \frac{3}{4} \sum_i \log \alpha_i + O_\ell \left(\frac{\log \log A}{\log A} + \frac{(\log A)^2}{A} \right).$$

Replacing $\sqrt{Ay_a}$ on the left by $\sqrt{Ab} u_{b,a}$ incurs an additional $O_\ell(\log A \sqrt{A/b})$ error.

Proof. The Laguerre–Bessel inequality, uniformly when $y \rightarrow 0$, gives $H(y) = y^\ell \prod_i \Phi(a_i y)(1 + O_\ell(y))$. Write $x = \sqrt{Ay}$. Bracketing first at fixed small and large multiples of $\log A$ shows that the root has $x \asymp \log A$. The standard large-argument expansion of I_1 then gives uniformly over the stated proportions

$$H(x^2/A) = \frac{x^{\ell/2} e^{2hx}}{A^\ell (4\pi)^{\ell/2} (\prod_i \alpha_i)^{3/4}} (1 + O_\ell(x^{-1} + x^2/A)).$$

Taking logarithms, substituting $x = (\ell/(2h)) \log A + O_\ell(\log \log A)$ in the logarithmic term, and using Theorem 10.3 proves the claims. The constants also depend on the fixed compact set of proportions. $\square$

Proposition 10.5 (probability approximation can fail while roots agree). *Let $a = \lfloor b/\log b \rfloor$, let y_a solve $B_a(y_a) = 1$, and set $u_0 = \sqrt{y_a/b}$. Then*

$$H(y_a) = B_a(y_a)^2 = 1, \quad S_{(b,a,a)}(u_0) \rightarrow \infty, \quad \frac{u_{b,a,a}}{u_0} \rightarrow 1.$$

Thus crossing roots admit a uniform approximation in a regime where the corresponding relative probability approximation fails: $S(u) \sim H(bu^2)$ is false as a uniform assertion over $A = o(b)$.

Proof. Retain only run words that start and end in the large color and have exactly one adjacent pair of distinct rare colors. The run enumeration in Theorem 9.6 shows that their contribution is the polynomial $2uyB'_a(y)^2$, with each term of total rare-run count m additionally multiplied by $\prod_{j=1}^{m-1} (1 - j/b)_+$. Give J probabilities proportional to $\binom{a-1}{j} y^j/j!$, the coefficients of $B'_a(y)$. Its differential equation $y(B'_a)'' + (1+y)(B'_a)' - (a-1)B'_a = 0$ implies $\mathbb{E}J^2 + y\mathbb{E}J = (a-1)y$; hence $\mathbb{E}J \leq \sqrt{ay}$ and $\mathbb{E}J^2 \leq ay$. For two independent copies, $m = J_1 + J_2 + 2$, so

$$\mathbb{E}[m(m-1)] \leq 4ay + 6\sqrt{ay} + 2.$$

The clipped-product deficit bound therefore proves

$$S_{(b,a,a)}(u_0) \geq \frac{2\kappa_a(y_a)^2}{bu_0} \left(1 - \frac{4ay_a + 6\sqrt{ay_a} + 2}{2b} \right).$$

The preceding corollary for one rare color gives $\sqrt{ay_a} \sim \frac{1}{2} \log a$, and the elasticity bounds give $\kappa_a(y_a) \sim \sqrt{ay_a}$. The last display is asymptotic to a lower bound $\sqrt{a/b} \log a \sim \sqrt{\log b}$, which diverges. The root assertion follows directly from Theorem 10.3, since $2a = o(b)$. $\square$

These results complete the leading near-vertex threshold for any fixed number of rare colors. They do not classify every finite-size global mode, nor do they assert a uniform higher-order root expansion for all growing rare counts. The analytic arguments in this section have written proofs and are not included in the finite Lean formalization.

11 Spatial superlevel sets and weak-field face phases

The interior-composition estimate also determines the geometry of the bins whose individual probabilities exceed the vertex probability. We give the result in a weak external field, since this perturbation exhibits additional face phases that the symmetric critical window suppresses.

Fix d and the symmetric transition model of Section 8. For $h = (h_1, \dots, h_d) \in \mathbb{R}^d$, define the *tilted occupation law*

$$\tilde{P}_{N,h}(n) = \frac{e^{h \cdot n/N} P_N(n)}{\sum_{m_1 + \dots + m_d = N} e^{h \cdot m/N} P_N(m)}. \quad (59)$$

Thus the field on each step is h_i/N . This is a Gibbs tilt of the occupation law; it is not a claim that the original homogeneous transition probabilities have acquired this bias. The normalization cancels in every bin comparison. A modal phase below specifies which face contains the largest individual bin probabilities; it does not assert concentration of the total probability mass on that face. Put $L = \log N$, retain

$$u_N(t) = \frac{L - \frac{1}{2} \log L + t}{N}, \quad p_N(t) = \frac{1}{1 + (d-1)u_N(t)},$$

and write $h_* = \max_i h_i$ and $\tilde{V}_{N,h} = \max_i \tilde{P}_{N,h}(N e_i)$. For a support $S \subseteq \{1, \dots, d\}$ of cardinality $k \geq 2$, set

$$\bar{h}_S = \frac{1}{k} \sum_{i \in S} h_i, \quad B_S(t, h) = (k-1)(t - a_k) + \bar{h}_S - h_*, \quad H_S = \{y \in \mathbb{R}^S : \sum_{i \in S} y_i = 0\}.$$

Euclidean measure on H_S is understood, and $\omega_s = \pi^{s/2}/\Gamma(1 + s/2)$ denotes the volume of its s -dimensional unit ball when $s = k-1$.

Theorem 11.1 (Gaussian superlevel geometry). *For a count vector with support exactly S , put*

$$y_i = \sqrt{L} \left(\frac{n_i}{N} - \frac{1}{k} \right), \quad i \in S.$$

Uniformly when t, h, y range over fixed compact sets, with $y \in H_S$,

$$\frac{\tilde{P}_{N,h}(n)}{\tilde{V}_{N,h}} = \exp \left(B_S(t, h) - \frac{k^2}{4} \sum_{i \in S} y_i^2 + o(1) \right). \quad (60)$$

Let $M_{N,S}(t, h)$ be the number of bins of support exactly S whose tilted probability is strictly greater than $\tilde{V}_{N,h}$. Then, uniformly for (t, h) in compact sets,

$$\frac{L^{(k-1)/2}}{N^{k-1}} M_{N,S}(t, h) \longrightarrow \frac{\omega_{k-1}}{\sqrt{k}} \left(\frac{4[B_S(t, h)]_+}{k^2} \right)^{(k-1)/2}, \quad [x]_+ = \max(x, 0). \quad (61)$$

If $B_S(t, h) < 0$, there are no such bins for all sufficiently large N . At $B_S(t, h) = 0$, the normalized count tends to zero; no eventual emptiness is asserted at that boundary.

Proof. With $\alpha_i = 1/k + y_i/\sqrt{L}$ and $\sum_i y_i = 0$, Taylor expansion on any fixed compact set in H_S gives

$$\left(\sum_i \sqrt{\alpha_i}\right)^2 = k - \frac{k^2}{4L} \sum_i y_i^2 + O(L^{-3/2}), \quad D(\alpha) = D_k(1 + o(1)).$$

Insert these expressions and $z = L - \frac{1}{2} \log L + t$ into Theorem 8.5. Since $\log D_k = -(k-1)a_k$, the logarithm of the untilted ratio is

$$(k-1)(t - a_k) - \frac{k^2}{4} \sum_i y_i^2 + o(1).$$

The logarithm of the field factor relative to the largest vertex is $h \cdot \alpha - h_* = \bar{h}_S - h_* + o(1)$, uniformly on the stated compact sets. This proves (60).

The local estimate must also exclude superlevel bins near the boundary of the face. We supply that step rather than assuming that a local limit controls all bins. Fix compact sets of t, h , and choose a sufficiently large constant R . Starting from any positive count vector on S , successively transfer one unit from a coordinate b to a coordinate a when $b \geq a + 2$. These transfers preserve the support and terminate at a balanced vector. By Theorem 8.6, the untilted probability strictly increases at every transfer. Each transfer changes the scaled vector y by norm $\sqrt{2L}/N$ and decreases its squared norm. If the starting vector has $\|y\| > R$, the first vector along this path with norm at most R has norm $R + o(1)$; it lies in a compact subset of the relative interior of the face. Its untilted log ratio to a vertex is at most

$$(k-1)(t - a_k) - k^2 R^2/4 + o(1).$$

This is an upper bound for the starting vector's untilted log ratio. For every composition α , $h \cdot \alpha - h_* \leq 0$. Consequently R can be chosen uniformly so that every tilted superlevel bin satisfies $\|y\| \leq R$. The same argument, with a larger R if necessary, bounds all face maximizers when comparison is made with its balanced bin.

Inside this ball the logarithms in (60) converge uniformly. For any $\delta > 0$, their positive set is sandwiched between the lattice points in balls of squared radii $4[B_S - \delta]_+/k^2$ and $4[B_S + \delta]_+/k^2$, with the inner ball interpreted as empty when $B_S \leq \delta$. The scaled count vectors form a translate of the lattice

$$\frac{\sqrt{L}}{N} \{z \in \mathbb{Z}^S : \sum_i z_i = 0\} \quad \text{in } H_S.$$

Its covolume is $\sqrt{k}(\sqrt{L}/N)^{k-1}$: the basis $e_i - e_k$, $1 \leq i < k$, has Gram determinant k . Lattice Riemann sums over balls therefore give the stated volume constant. Letting $\delta \downarrow 0$ proves (61), including $B_S = 0$. The radii stay bounded on compact parameter sets, so the same sandwich proves uniform convergence. If $B_S < 0$, the uniform profile on the confining ball is strictly below one for large N , proving emptiness. $\square$

At zero field, all supports of the same size have the same count, so the right-hand side of (61) is multiplied by $\binom{d}{k}$ when counting all bins with k positive coordinates. For $k = 2$, writing $y = (y_1, -y_1)$ gives the profile $\exp(t - a_2 - 2y_1^2)$ and the one-face count constant $\sqrt{2[t - a_2]_+}$, agreeing with the free binary window after its centering convention is taken into account.

Theorem 11.2 (weak-field modal phase diagram). *Define the face scores*

$$\lambda_S(t, h) = \begin{cases} h_i, & S = \{i\}, \\ (|S| - 1)(t - a_{|S|}) + \bar{h}_S, & |S| \geq 2. \end{cases}$$

The largest bin weight, expressed relative to the untilted vertex before normalizing (59), satisfies

$$\log \max_n \left\{ e^{h \cdot n/N} \frac{P_N(n)}{V_N} \right\} \longrightarrow \max_{\emptyset \neq S \subseteq \{1, \dots, d\}} \lambda_S(t, h). \quad (62)$$

For fixed (t, h) , every global mode has, for all sufficiently large N , a support maximizing the limiting score. If the maximizing support is unique, all global modes have exactly that support. On any selected support S with $|S| \geq 2$, their locations satisfy

$$\max_{i \in S} \left| \frac{n_i}{N} - \frac{1}{|S|} \right| = o((\log N)^{-1/2}).$$

These assertions do not identify the maximizing integer vectors exactly, and equality of limiting scores does not assert finite- N coexistence.

If the fields are sorted $h_1 \geq \dots \geq h_d$, the score maximization reduces to the upper envelope of the d lines

$$\lambda_1(t) = h_1, \quad \lambda_k(t) = (k - 1)(t - a_k) + \frac{1}{k} \sum_{i=1}^k h_i, \quad 2 \leq k \leq d. \quad (63)$$

Away from ties, the winning support size is nondecreasing as t increases. If the fields are strictly ordered, the corresponding supports are the nested sets $\{1, \dots, k\}$. Every support size $k \in \{2, \dots, d - 1\}$ can be the unique winning size on an open interval for suitable fixed fields. Thus proper-face phases can persist asymptotically under a field of order $1/N$ per step.

Proof. For each fixed support, the confinement argument in the preceding proof applies also to its maximum, because the field factor is bounded above and below uniformly in N . On that compact scaled region, the profile relative to the untilted vertex is $\exp(\lambda_S - k^2 \|y\|^2/4 + o(1))$. Balanced integer vectors have $y = o(1)$, so the face maximum converges to e^{λ_S} . There are finitely many supports. Taking their maximum proves (62), and the positive gap to any nonmaximizing score excludes its support for large N . If a sequence of maximizers within a selected face had scaled norm bounded away from zero, the uniform Gaussian profile would make it strictly smaller than that face's balanced bin. This proves the location statement.

Among supports of a given size k , the largest field average is the average of the k largest fields. This proves (63). For $t_1 < t_2$, a line of strictly larger slope cannot win at t_1 and lose to a line of smaller slope at t_2 : subtracting the two comparisons would give $(k_1 - k_2)(t_2 - t_1) \leq 0$. Hence the winning size is nondecreasing away from ties.

To realize a prescribed $2 \leq k < d$, set $h_1 = \dots = h_k = 0$ and $h_{k+1} = \dots = h_d = -H$. Choose $t > a_k$ sufficiently close to a_k so that $t < a_{k-1}$ if $k > 2$. The k -face score is positive; all smaller nonvertex face scores are negative, and the largest vertex score is zero. For $j > k$, the largest score is

$$(j - 1)(t - a_j) - H \frac{j - k}{j}.$$

Taking H sufficiently large makes each of these strictly smaller than the k -face score. Every other support of size k includes a negative field and also loses. All comparisons are strict, so they persist on an open interval of t . Small perturbations can make the fields strictly ordered without removing the phase. $\square$

Corollary 11.3 (one field realizes the full face hierarchy). *Fix $d \geq 2$ and $H > 3/4$, and choose*

$$h_i = -H(i-1)^2, \quad 1 \leq i \leq d.$$

In particular $h_i = -(i-1)^2$ is a single explicit choice valid in every fixed dimension. Put

$$c_k = (k + \frac{1}{2}) \log k - \frac{k-1}{2} \log(4\pi), \quad \tau_k = c_k - c_{k+1} + \frac{H(4k-1)}{6} \quad (1 \leq k < d),$$

and set $\tau_0 = -\infty$, $\tau_d = +\infty$. Then $\tau_1 < \dots < \tau_{d-1}$, and for every $k \in \{1, \dots, d\}$ the support $\{1, \dots, k\}$ is the unique winning support throughout the nonempty interval $\tau_{k-1} < t < \tau_k$. Consequently, for each fixed t in this interval and all sufficiently large N , every global mode of the tilted law has exactly those k positive coordinates.

Proof. Here $c_1 = 0$ and $c_k = -(k-1)a_k$ for $k \geq 2$. Since $k^{-1} \sum_{i=1}^k (i-1)^2 = (k-1)(2k-1)/6$, the lines in (63) have intercepts $c_k - H(k-1)(2k-1)/6$. The intersection of lines k and $k+1$ is therefore $t = \tau_k$. Extend c_k to the displayed smooth function $c(x)$ for $x \geq 1$. Direct differentiation gives

$$0 < c''(x) = \frac{x - 1/2}{x^2} \leq \frac{1}{2}.$$

Integrating this bound twice over unit intervals yields $c_{k+2} - 2c_{k+1} + c_k \leq 1/2$. Hence

$$\tau_{k+1} - \tau_k = \frac{2H}{3} - (c_{k+2} - 2c_{k+1} + c_k) > 0.$$

For $\tau_{k-1} < t < \tau_k$, summing consecutive differences $\lambda_{j+1}(t) - \lambda_j(t) = t - \tau_j$ shows that line k strictly exceeds every earlier and later line. The fields are strictly ordered, so its unique best support is $\{1, \dots, k\}$. Apply Theorem 11.2 to obtain the finite- N conclusion. $\square$

Example 11.4 (a persistent triangular edge phase). For $d = 3$ and $h = (0, 0, -H)$, the three competing maximal scores are

$$0, \quad t - a_2, \quad 2(t - a_3) - H/3.$$

If $H > 6(a_2 - a_3)$, the edge with support $\{1, 2\}$ wins strictly for

$$a_2 < t < H/3 + 2a_3 - a_2.$$

For every fixed t in this interval, all global modes of the tilted law lie on that edge for sufficiently large N . This differs from the untilted law, where proper-face modes disappear from the asymptotic critical window. It does not strengthen the finite-size counterexample of Example 8.8 into an untilted asymptotic claim.

12 Computational verification

The script `verify.py` uses exact rational arithmetic except where it says otherwise. It computes P_N in three ways: the closed form of Theorem 2.1, the recursion (2), and enumeration of all 2^N words weighted by (1). The three agree, and rows sum to 1, for $N \leq 16$ and 20 values of p ; since $P_N(k)$ is a polynomial of degree less than N in p , this is an identity in p . The closed form and the recursion also agree for $N \leq 40$. The numerators reproduce rows 1 to 9 of the three tables in Kagey's figure. The generating functions (3) and of Proposition 3.1 and the recurrence (4) hold

to degree 16, and Proposition 2.3 is checked against enumeration for $d = 2$ ($N \leq 12$) and $d = 3$ ($N \leq 8$). The $p = \frac{2}{3}$ numerators equal the 60 published terms of A035002 and a direct count of rook paths for $N \leq 24$; the $p = \frac{1}{3}$ numerators equal the 55 published terms of A348595 and a direct count of the blocked walks for $n, k \leq 12$. Proposition 4.1 agrees exactly with the moments of the computed distribution for $N \leq 16$ and $N = 25, 40$, and Remark 4.4 with enumeration of all 3^N words for $N \leq 9$. In floating point, the eigenvalue formula for φ_N agrees with the computed distribution to 10^{-12} , and at $N = 2000$ the distribution function of K_N , computed by (2) and again by (4), is within 1.1×10^{-4} of the normal one (with continuity correction) for $p = \frac{1}{3}, \frac{2}{3}, \frac{5}{6}$.

For Theorem 5.1 the script checks the polynomials in (b) exactly, the coefficient inequality for $N \leq 300$, the single sign change of $P_N(m_N) - P_N(0)$ on the grid $p = i/200$ for $N \leq 14$, and the monotonicity of p_N for $N \leq 121$. Each entry of Table 1 is the root of S_N found by bisection in integer arithmetic and confirmed by evaluating (2) exactly at the two ten-digit neighbours of the root. The roots at $N = 800, 1600, 3200$ are confirmed by the floating-point recursion, and x_B is computed by power series and again by quadrature. The run takes between one and two minutes.

The supplementary script `check.bessel.py` checks the uniform bound on 1,400 even/odd test cases and the quantitative root bound through $N = 10^8$, using 80-digit decimal arithmetic. These are numerical corroborations, not interval-certified computations. The script `verify-geometry.py` compares 6,733 exact rational bin probabilities in 237 distributions with an independent dynamic program; 158 distributions also agree with exhaustive word enumeration. It verifies normalization, nonnegativity, means, covariances and endpoint cases for $d = 2, 3, 4$. The script `verify_adjacent.py` checks the adjacent-bin polynomial and the consequences for global extrema. All checks can be run with `python verify_all.py`.

The additional script `verify_allbin.py` checks 5,476 coefficient comparisons exactly, 264 all-bin Bessel inequalities, 28 Laguerre–Bessel inequalities, and crossing expansions with 80-digit arithmetic, including highly unbalanced bins. The script `verify_periodic_window.py` checks 66 cyclic polynomials against exhaustive word enumeration, 120 periodic relative-error inequalities, and root/window samples through $N = 10^{12}$. The script `verify_simplex_threshold.py` checks the positive refresh formula against enumeration, strict balancing transfers, the small edge-mode counterexample, and face-center crossing expansions. Numerical asymptotic checks are corroboration, not interval certificates or substitutes for the analytic proofs. The scripts `verify_frontier_boundary.py` and `verify_frontier_modes.py` check the added uniform estimate, rare-coordinate asymptotics, spatial profile and weak-field phase assertions against finite calculations. Detailed records are saved in the accompanying JSON files under `data/`. The script `verify_vertex_completion.py` adds 462 exact identity comparisons, 200 rational elasticity checks, 16 explicit root-bracket checks, and four high-precision growing-rare-count stress cases. The runner now includes ten scripts. These finite checks do not formalize the uniform analytic proof.

The `lean/` directory contains separately compiled Lean proofs of finite product and algebraic ingredients. It does not formalize the infinite Bessel sums, analytic inversion, Laplace expansion, or complete asymptotic theorems. Its coverage note and compiler/axiom logs give the exact formal statements checked.

13 Further questions and scope

The original distribution, lattice-walk and central-crossing questions are answered, as are the cylinder pushforward and the explicitly specified symmetric tetrahedral model. The extensions give quantitative thresholds for all binary bin distances and all fixed simplex face dimensions, together with free/periodic boundary comparisons and critical mode windows. The extensions also

supply a composition-uniform relative bound, mixed-face boundary crossovers, spatial superlevel volumes and weak-field face phases. The leading near-vertex crossing is now uniform for every fixed number of rare colors with total $A = o(b)$. Uniform higher-order corrections in that whole regime, and a growing number of colors, are not established here. Their proposed novelty lies in these refinements, subject to the prior-art qualifications above. A complete multistate finite-size mode diagram outside the proved window remains open here, as do nonuniform switching and alternative geometric rules. The finite-size edge-mode example prevents promoting the asymptotic vertex/center dichotomy to an unrestricted theorem.

Data and code availability. The verification scripts (plain Python, standard library only) and prediction ledger accompany this document. The ledger preserves the original predictions and adds the later proof status. The repository’s `PUBLICATION.md` records the separate literature and venue assessment; it is not a claim of acceptance or novelty.

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